

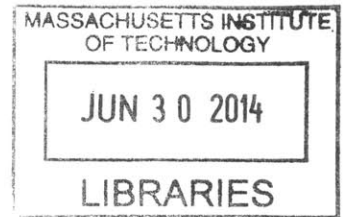
DESIGN CONSIDERATIONS FOR GE-ON-SI WAVEGUIDE PHOTODETECTOR

by

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B.S. Electrical Engineering,
University of Belgrade, Serbia (2011)

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Submitted to the Department of Electrical Engineering and Computer
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Abstract

In integrated photonic circuits photodetector is one of key components, modern applications require that photodetector has a high 3 dB bandwidth. The ultimate limit for the response time for conventional photodetectors (like vertically illuminated photodiode, Schotky photodiode, MSM photodetector etc.) is given by the transit time of the photogenerated electron-hole pairs, it can not be minimised by decreasing the thickness of the depletion region without reducing quantum efficiency (i.e. the fraction of the incident light that is absorbed). Waveguide photodetectors have been developed to overcome this trade-off. In the waveguide photodetector light propagates in a direction that is parallel to the junction interfaces and is perpendicular to the drift of the generated electron-hole pairs. This geometry decouples absorption length from the drift length. Therefore the waveguide photodetector can have both a very thin active region for short transit time and a long absorption length for a high quantum efficiency. In this thesis, I designed germanium on silicon photodetector. The main designing tool was full vectorial 3D Finite Difference Time Domain (FDTD) simulator. Bandwidth-efficiency product was used as the main figure of merit. The input is silicon rib waveguide, which is optimised to maximize transmitted power. For optimal dimensions of the device calculated responsivity is 0.94 A/W, efficiency is 83 %, bandwidth is 64 GHz and bandwidth $\times$ efficiency product is 53 GHz.

Thesis Supervisor: Franz X. Kärtner

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Chapter 1

Introduction

1.1 A Historical Perspective

In 1965, Gordon Moore published a well-known paper “Cramming more components onto integrated circuits” [1]. In this paper, he described the trend and also predicted the future of integrated circuits by observing the fact that the number of transistors on integrated circuit doubles approximately every two years – known as Moore’s law. Indeed, over the period of four decades, key features like processor speed and memory size are roughly doubling each 18 months. This steady development is the very base of the Information and Communications Technologies (ICT) market today and, as such, it is strongly linked to the dynamics of micro-electronic integration technology.

Electronic integrated circuits provide many functions in optical networks, for example. Some of these functions include monitoring data transmission performance, tracking service level agreements, providing fault detection, protection against service outages, switching different data streams into

larger transmission facilities etc. Even though optical networks manage photons, and not electrons, majority of value-added service functionality is provided by combined use of electronic ICs and system software. Purely optical technologies (Wavelength Division Multiplexing (WDM), optical amplifiers etc.) are reserved mostly for enabling capacity scalability and extending optical transmission between nodes of a network.

Benefits provided by electronic ICs cost tens to hundreds of dollars per IC. In contrast, optics-based technologies are more expensive, more complex, or deliver less functionality. This difference in cost of implementation yields the fundamental lead of electronic over optical solutions. However, the cost of electronic ICs is not always the issue. Much effort has been put into the development of “all-optical” networks that seek to minimize Optical-to-Electrical-to-Optical (OEO) conversions. In such network, electronic processing is assigned to edges of the network, while service manipulation within the core belongs to photonic domain. Cost of this conversion between optical and electronic domain is the price that is essentially paid when it comes to use of electronic ICs.

OEO conversions are expensive because of the everlasting need of each conversion for additional single-use, individually packaged devices: lasers, modulators, wavelength lockers, detectors, WDM multiplexers and demultiplexers, attenuators. Sometimes a single OEO conversion can require up to half a dozen additional optoelectronic components. Since the conversion cost of transferring data between these two domains is so high, the main benefit of electronic ICs – very low cost per device – is often overridden, Figure 1.1 .

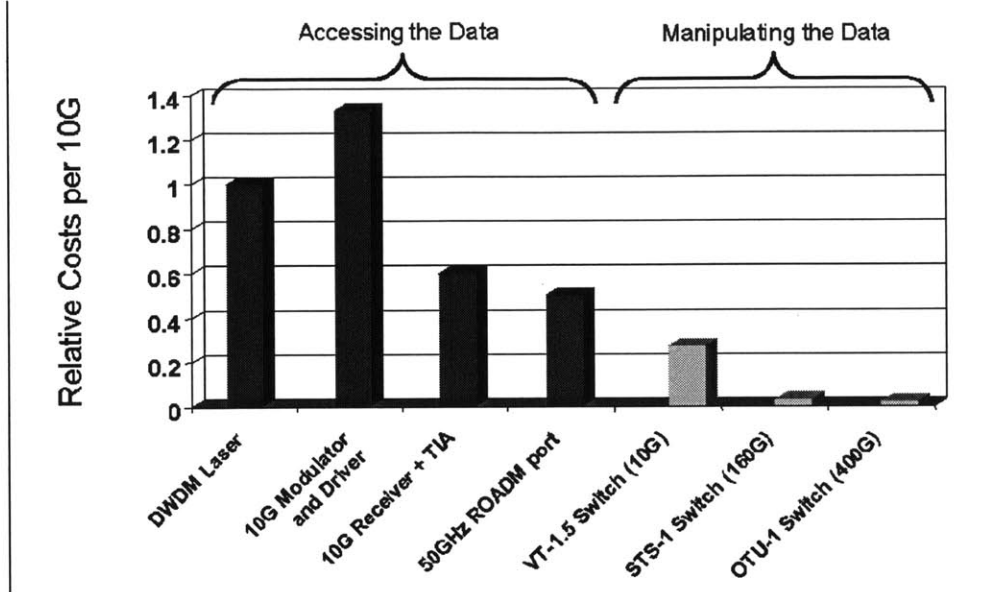


Figure 1.1: The cost of optical components required to implement an OEO conversion are significant compared to the cost of electronic ICs used to manipulation the data in electronic domain [2]

Solution to this problem lies in use of Photonic Integrated Circuit (PIC), which is conceptually very similar to an electronic IC. By definition an integrated circuit is a micro-electronic device that houses multiple circuits on a chip. For example, an IC is built by lithographic fabrication of numerous transistors on a silicon chip. Similarly, a photonic IC is a device that houses integrated photonic functions on a chip. PIC perfectly unifies all individually packaged additional components required for OEO conversions into a single device. This increases efficiency by eliminating the need to separately fabricate, package, burn-in and test all of the discrete single-function devices. Consolidating these devices into a single PIC leads to higher architectural sustainability, cost, power and reliability advantages. According to A. Rah-

man [3], PICs technology must meet to following criteria:

- it must be capable of creating a broad range of optical functions out of a single fab or process;
- the means must exist for it to be readily manufactured at low cost in high volume;
- the capability must be developed to aggregate individual optical functions into more complex arrangements within the technology and with other optical technologies.

The first step in realizing a PIC device is a design of the structure with desired optical properties, which is usually accomplished by combination of simulation and prototyping. Once a design is finalized, various lithography techniques, including reactive ion etching and e-beam lithography, can be used to fabricate devices in batch. Paralleling the revolution of microelectronics, both monolithic and hybrid approach remain available in photonic integration as well. In a hybrid PIC, multiple discrete optical devices are gathered into a single housing, sometimes with associated ICs, and inter-connected to each other. This can be rather complicated to produce, as many single-use devices must be inter-connected internal to package with sub-micron tolerance. Also, differences in optical, mechanical and thermal characteristics of various materials must be well coordinated. Monolithic integration, on the other hand, consolidates many devices into a single photonic material, thus providing us with the greatest level of simplicity and reliability benefits. Multiple optical components are built into a common substrate and form a single, physically unique device. These components include lasers,

modulators, attenuators, multiplexers and demultiplexers, optical amplifiers, couplers, filters – for each of these devices a broad variety of different operation principles and materials has been reported. Naturally, realizing even a modest subset of these devices in monolithically integrated technology is a scheme of great magnitude. According to some authors, the key to a successful photonic integration is to reduce this variety of optical functionalities to a few elementary components with broad application possibilities. Smit et al. [4] suggest three elementary devices:

- a passive waveguide structure that allows low-loss interconnection of devices and realization of miniaturized components like couplers, filters, multiplexers, polarization and mode converters;
- an element for manipulating the phase of optical signals – such choice has been made for a fast electro-refractive modulator; main applications are fast optical switches and modulator, for both phase and amplitude;
- an element for manipulating the amplitude of optical signals – the Semiconductor Optical Amplifier (SOA), which modulated the phase and allows for both linear and non-linear signal processing (WDM light sources, femtosecond pulse lasers, ultrafast optical switches).

In contrast to solution proposed by Smit et al. [4], some authors, like A. Rahman [3], envisioned an integrated monolithic design where a single mask is used to layout all components, including waveguide interconnects. This results in true PIC that could pave the road for next generation of fiberoptic communication, computing, sensing.

One of the major challenges of photonic integration is the proper choice of the substrate material. For a truly integrated photonic technology, a smart material system is necessary: the one that can function in a similar fashion as silicon in IC technology. Nowadays, optical components are built from Indium Phosphide (InP), Gallium Arsenide (GaAs), Lithium Niobate (LiNbO_3), Silicon (Si), Silicon-on-Silica. It must be noted here that photonic integration derives its value from ability to incorporate as many disparate functions as possible into a single material platform.

Since devices can be monolithically interconnected by on-chip waveguides, InP based PICs enable the fabrication of system-on-a-chip or an “optical processor”. This can provide substantial benefits versus the use of discrete devices. Indium Phosphide is ideal material for implementing large scale monolithically integrated PICs because it supports the integration of almost all function required in ICT applications: light generation, amplification, modulation and detection.

In order to acquire and keep important role in the ICT market, photonics must also obey Moore’s law. Objective remains the same as in micro-electronics: to reduce device dimensions and fabrication costs over a longer period of time. The question is: does micro-photonics technology have the same potential as micro-electronics technology to reach this objective? Smit et al. [4] were one of the first authors who have studied this question and interpreted the answer visually, Figure 1.2. They have put the development of PIC in their own lab in a graph with the potential integration density in devices per square centimeter on the vertical axis.

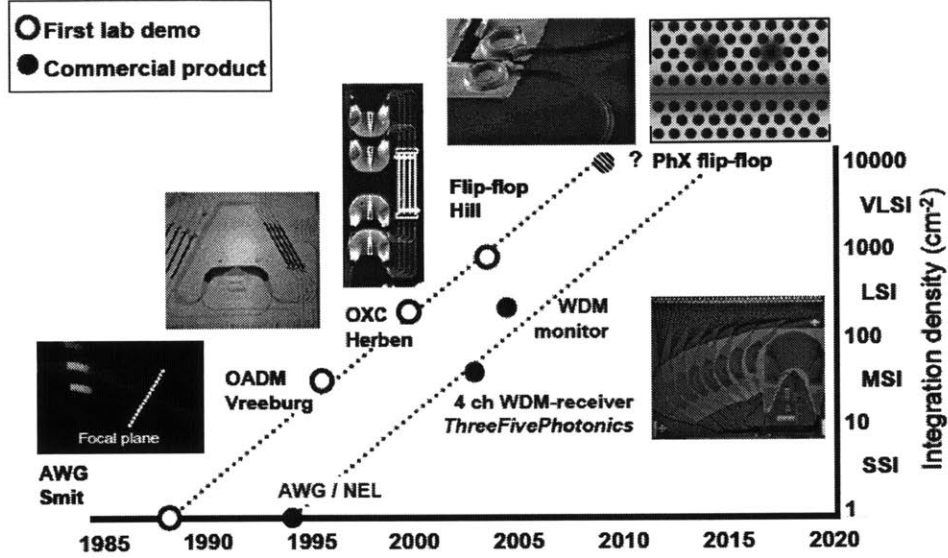


Figure 1.2: Moore's law in micro-photonics [4]

In this graph, the open circles mark the first publication of a device or a circuit. It starts with the invention of the arrayed waveguide grating (AWG) in 1988. The next circle is the first InP-based Optical Add-drop Multiplexer (OADMP) in 1997, a device that integrated a single arrayed waveguide gratings (AWG) with four Mach-Zender switches on an area of 0.2 cm^2 ; 25 components per square centimeter. Smit et al. [4] also developed a technology for reducing the size of their AWG's using deep etching technology, hence producing the world's most compact Optical Cross-Connect (OXC). This is the device with an integration density with an integration density with more than 100 components per square centimeter, featuring 4 AWG's and 4 Mach-Zender switches on an area of 5 mm^2 . Further reduction of AWG-size is reached, approaching to the limits of conventional deep etched waveguide

technology in InP: $250 \times 350 \mu\text{m}^2$. The last circle in this figure represents a photonic flip-flop, which consists of two deep-etched micro-ring lasers, published by Hill et al. [7]. Dimensions of this device are $20 \times 40 \mu\text{m}^2$ with integration density of more than 1000 components per square centimeter.

Along with these devices, some commercial products are also placed on the graph. The first one is NEL's AWG, dating back from 1994. The second point is a WDM receiver, brought to market by ASIP in 2003. This device consists of one AWG and four detectors. The third point is a WDM Channel monitor, made by ThreeFivePhotonics in 2004 and it consists of 9 AWG's and 40 detectors. Arranged in a graph as described, these devices together fit to a straight line with a slope slightly larger than Moore's law. With this sample, Smit et al. [4] validated Moore's law in microphotonic integration technology.

As can be observed, the integration scale in PIC is being shifted to VLSI-level and technologies for reduction of device dimensions are creditable for that fact. We already mentioned deep waveguide etching technology, which causes a strong lateral confinement of light. Narrower size and much lower bending loss are just few of the many advantages of deeply etched waveguides in compare to shallow etched ones. Key components in PIC can be made much smaller this way, with significantly increased functionality of components as such. This will eventually bring us to the fundamental limits of photonic technology by ongoing scaling of device capacity and functionality. Moore's law, of course, will continue to provide the dynamics of such development.

Obviously, foundation for rapid progress in PIC's has been laid. Routine

fabrication of grating-based DFB and DBR lasers allowed high-Q on-chip resonators without cleaved facets. High-quality computer-automated vapor and beam growth systems, such as metal-organic vapor-phase (MOVPE) for the InP-based materials enabled the fabrication of quantum-well lasers in the InGaAsP system. These systems have offered the reproducible growth of highly complex vertical layer structures, with large number of ultrathin etch-stop layers included. Advances like this one brought a freedom in the design of integrated waveguide devices. Another advance worth mentioning here is MOVPE's capability for good Fe-doped InP regrowths in various geometries, which led to new PIC processing techniques where each device maintained a high degree of optimization in its layer structure and geometry.

Throughout development, design and fabrication of PICs, various problems emerge. These issues can be crudely divided into three groups: optical engineering problems, optoelectronic-electronic engineering problems and electrical problems.

Optical problems include, among other:

- fabrication of low-loss optical waveguides with the associated constraints on doping types and levels;
- design and fabrication of low-loss longitudinal coupling between active and passive portion of devices;
- improved coupling from external fiber sources into the tight waveguides, for optimization purposes;
- requirement of strictly single-mode guides in the passive structures.

Some of optoelectronic-electronic problems are:

- requirement for current blocking in lasers, and current of field access to any active sections in the PIC, while retaining low-doped or Fe-doped InP cladding for the low-loss waveguide interconnections;
- requirement for high electrical isolation between the various active devices in a PIC, which is essential to avoiding crosstalk in multichannel PICs.

Electrical problems are usually encountered in contacting or mounting PICs, where inductive or capacitive coupling may occur in high-speed applications. These three problem categories must be addressed in order to properly execute design and fabrication of PICs. That being said, it is crucial to avoid unnecessary complications in crystal growth or fabrication-processing while solving these problems.

As outlined before, electronic ICs and PICs are conceptually similar: photonic waveguide is somewhat analogous to transistor. Just as transistor is the basic building block of electronic IC, so is photonic waveguide the building block of PIC. Although physics of photon differs greatly from the physics of electron, one could say that waveguide processes optical signal similarly as transistor processes electronic signal. Waveguide can be designed to perform number of optical functions we mentioned (amplification, modulation, switching etc.) and number of photonic devices featuring this waveguide can be designed to carry out various photonic signal processing. Sometimes even a relatively simple assembly of waveguides in a form of grating accomplishes a PIC. This is a PIC with common application – the wavelength division

multiplexing (WDM) and demultiplexing on a chip, commonly known as arrayed waveguide grating (AWG).

For a photonic waveguide to act more like a transistor, it needs to both guide and amplify the photons and also modulate photonic signals on the same chip. In order to accomplish this by means of monolithic integration, a specific material system is needed – the one that can be processed synergistically without requiring multiple processes at each step. Silicon provides quality waveguiding and amplifying, but it is a poor choice of material when it comes to modulation, due to its' indirect bandgap and poor electro-optic properties.

Basic fiberoptic infrastructures still rely heavily on silicon devices. Silicon is extremely matured in terms of processing, lending means to integrate CMOS processes and photonic functions on a single chip. Silicon integrated photonics has many merits. It can be as fine as nanometer scale in structure, and as large as giga scale in complexity. Possibilities with geometry in silicon are endless. As the mainstream electronic devices are made of silicon, fabrication of photonic devices on silicon proves to be much cost effective method of integration. Indeed, the most advanced extension of a silicon photonics is to have a comprehensive set of optical and electronic functions available to the designer as monolithically integrated building blocks upon a single silicon substrate.

Within the range of infrared wavelengths, common to silica fiberoptic telecommunication systems ($1.3\ \mu\text{m}$ to $1.6\ \mu\text{m}$), silicon is transparent and generally does not interact with the light. This makes silicon exceptional medium for guiding optical data streams between active components. Active

devices such as light intensity modulator and photodetector can be created by incorporating additional materials (silicon dioxide, dopants, SiGe alloys) into design. However, due to indirect bandgap, low electro-optic and low non-linear coefficient, light emission from silicon is possible, but inefficient. This is one of the fundamental limitations of use of silicon in photonics, resulting in platforms that require light source as an external component. Full monolithic opto-electronic integration is a goal that is difficult to reach. Monolithic integration of electronics and optics is highly desirable – it reduces unwanted electrical parasitic and allows reduction in size. At Intel, two parallel approaches are currently being pursued:

- to achieve a high level of photonic integration with the goal of maximizing the level of optical functionality and optical performance;
- to look for specific cases where close integration of an optical component and an electronic circuit can improve overall system performance.

The latter led to integration of SiGe photodetector with a CMOS transimpedance amplifier. Intel is basically trying to find a way to *siliconize* photonics by making integrated photonic devices out of silicon instead of exotic material most manufacturers use today. This will remove a significant cost barrier in photonics and pave the way to producing photonics products based on silicon. Main source for achieving lower costs with higher performance (smaller size, lower power, higher data rate, greater transmit distance, expanded functionality, and expanded flexibility) in this field is the increase in optical complexity of the system. Some examples would be multiple wavelengths in one fiber from one ingress point, adaptive or re-

configurable optical components capable of recovering signal integrity under changing external conditions, all-optical packet switching, all-optical signal regeneration etc. This will, of course, require sophisticated electronic control solutions, which proves that monolithically integrated opto-electronic suite is natural progression of photonics industry.

Bulk silicon is an indirect bandgap material and cannot be efficient light emitter because the fast non-radiative recombination processes dominate the barrier transfer between the conduction and valence bands. As such, silicon is considered poor light emitter. Although, one might say that this situation is changing. Luminescence properties of silicon-based structures including porous silicon [12-15] and silicon nanoclusters in amorphous silicon-dioxide [16, 17] are being researched. Both red-orange band and blue band are observed in these structures. A green band has also been found in silicon nitride structure, providing the possibility for fabricating full-color devices based on silicon technology [18].

Obviously, there is ongoing effort to create a silicon-based emitter, but that work is still far from mature. Until an efficient silicon-based light source is available, a photonic integrated system will continue to use a conventional III-V material light emitter. Salib et al. [5] described a single mode, tunable external cavity laser (ECL), created by coupling an AR coated III-V semiconductor laser diode to a silicon-based waveguide Bragg grating. The lasing wavelength is selected by the grating and can be tuned by using the thermo-optic effect and simply heating the grating, producing a tuning rate of 12.5 nm/100° C. This way an inexpensive narrow line-width source can be produced and be suitable for optical communications.

The laser output carries no data or information, since it is a continuous wave. To encode data onto this continuous wave, an optical modulator is needed. Until recently, silicon optical modulators based on a waveguide could demonstrate barely moderate speeds of 20 MHz [20, 21]. Today's communication networks are demanding GHz performance, which means that devices from this category could spark no practical interest. Devices that have shown modulation frequencies in excess of 40 GHz [22, 23] are III-V semiconductor compounds and multiple quantum wells such as GaAs/Al-GaAs and InGaAsP-InP. These devices utilize the quantum confined Stark effect [24-26]. Although satisfying from the viewpoint of practical application, these devices are expensive to produce. The focus in research is moved to pursue for cost-effective silicon-based modulators with GHz performance. One of the cornerstone technologies in Intel is an experimental demonstration of a silicon optical intensity modulator with a modulation bandwidth of 2.5 GHz at optical wavelengths of around $1.55\ \mu\text{m}$, presented by Salib et al in [5]. This breakthrough happened by moving away from the conventional current injection-based design to a novel MOS capacitor-based architecture.

1.2 Motivation

The final optical component to be integrated onto an all-silicon optical platform is the photodetector. Silicon photodetectors for visible light ($0.4 - 0.7\ \mu\text{m}$) are widely used, because of their perfect efficiency at those wavelengths. However, silicon is naturally transparent in wavelengths typically used for optical communications ($1.31 - 1.55\ \mu\text{m}$), making the detection of

light in this range in silicon impossible. Pushing responsivity out to longer wavelengths could achieve efficient detection. This can be done by photodetectors based on SiGe alloys – a technology that is being developed in Intel [5]. Introducing Ge reduces the band gap and extends the maximum detectable wavelength. Figure 1.3 shows the effect on the absorption coefficient and penetration depth, defined as the distance that light travels before intensity falls to 36% ($1/e$). The data in this figure represents unstrained bulk material with no voltage applied. It is possible to shift the curves slightly to a higher wavelength when strain or electrical bias is introduced.

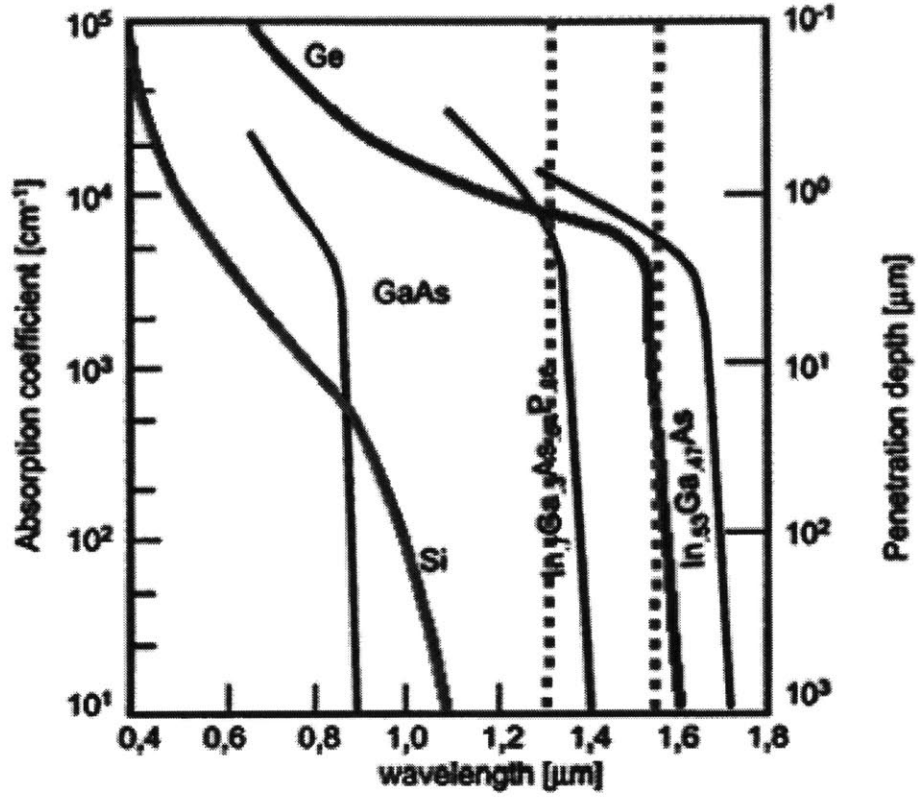


Figure 1.3: Absorption coefficient and penetration depth of various bulk materials as function of wavelength. The green lines mark the important wavelengths for telecommunications of 1.310 and 1.550 μm [5].

Two main benchmarks for a photodetectors are responsivity and bandwidth. Both of these are directly related to the absorption coefficient and penetration depth of the light. Responsivity is the ratio of collected photocurrent to the optical power incident on the detector. The bandwidth of a photodetector can be limited by the transit time required for the photocarriers to travel to the contacts or the RC time constant. One of the merits of

waveguide-based photodetectors is overcoming the inherent trade-off of photodetectors: maximizing the light absorption by making layers thicker results in a reduction of bandwidth due to transit time issues. When the light is incident from above and electrical and optical and electrical distances are coupled, one must choose between good bandwidth or high efficiency. However, by illuminating device from the side, photon-absorption path and a carrier-collection path are perpendicular to each other. This way the transit time can be kept low, while the effective length of the detector increases significantly. Another advantage of a waveguide detector is the planar nature of the device, which makes integration with other optical devices more accessible.

Figure 1.4 shows a cross-section of SiGe waveguide-based photodetector developed at Intel [5]. SOI platform is used as modulator; SiGe layer is directly on top of a silicon rib waveguide. Devices from this category achieved the responsivity as high as 0.1 A/W at 1.319 μm . This could be improved by a combination of increasing number of quantum wells used as absorbing material and changing the placement of SiGe in the waveguide. Altering the film composition could overcome limitation to bandwidth (< 500 MHz). Models predict data rates approaching to 10 Gb/s. The structure of this device is fully strained, preventing major defects in the active SiGe material.

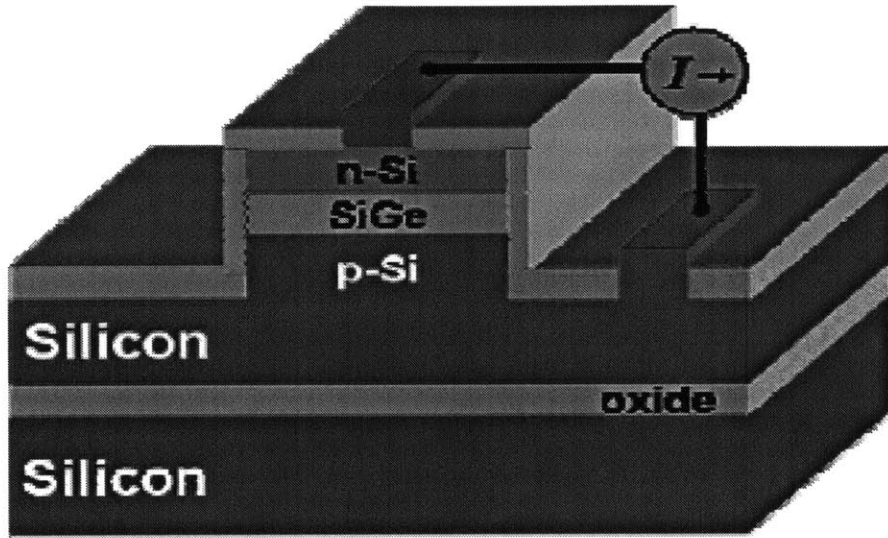


Figure 1.4: $\text{Si}_{1-x}\text{Ge}_x$ waveguide-based photodetector on SOI wafer. The waveguide is formed by the ridge of p -Si material and is running perpendicular to the cross-section. The SiGe MQW are inside the region labeled SiGe in the picture [5].

The amount of Ge required for efficient photodetection is dependent on the wavelength. For wavelengths used in optical communications, Ge concentration is needed to be over 40%. Major issues that rise in this integration are exposure to high temperatures after growth and chemical stability. Thus alternate processing modules must be developed in order to maintain the integrity of SiGe films.

Another interesting photodetector is the one reported by Ahn et al. [6]. This is a Ge p - i - n photodetector that is monolithically integrated with top coupled silicon oxynitride and silicon nitride waveguides, Figure 1.5. The

small size of the waveguide-integrated devices resulted in low absolute dark current. As previously mentioned, inherited efficiency-bandwidth trade-off is avoided, due to waveguide-based architecture of this device. This photodetector achieves the performance beyond the level possible with free-space illumination, especially at longer wavelengths where absorption in Ge is less efficient. High responsivity (~ 1.08 A/W) and high-speed (> 10 Gb/s) performance are obtained. The beauty of this device is that it retains its' high performance even at low operation voltages, thus satisfying the low-voltage requirement for CMOS circuits.

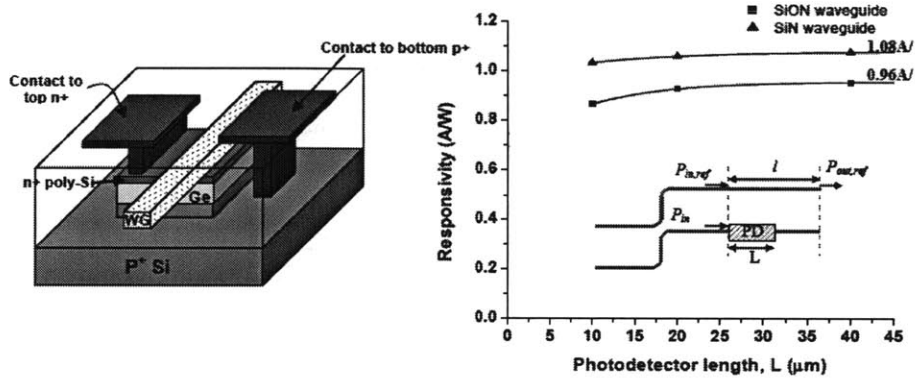


Figure 1.5: Schematic structure of a waveguide-integrated Ge *p-i-n* photodetector (left). The responsivity of waveguide-coupled Ge photodetector vs. detector length. An insert is the schematic layout of waveguide and photodetector devices on the chip [6].

In order to produce high-performance systems, one approach suggests to increase optical power incident on the wide-bandwidth photodetectors. This will allow the photogenerated RF output power (voltage swing) to directly drive the digital logic circuits, but higher performance photodiodes are required. Fundamentally, photodiodes are simple *p-n* junctions. The only

difference between the introductory device class junction and commercial products is extensive optimization. Namely, two factors limit a photodiode's output power: space-charge screening of intrinsic region electric field and thermal limitations. The latter is the result of the geometry and thermal conductivity of photodiode layers. At sufficiently high optical power levels, the space-charge induced electric field is strong enough to collapse the bias electric field. This results in loss of the RF signal. In traditional high-speed $p-i-n$ photodiodes, made of an InGaAs optically absorbing layer grown on InP substrate, composition and thickness of layers are chosen to balance trade-offs between power handling and frequency response. Tulchinsky and Williams [7] described a new photodiode structure, which uses a partially depleted absorbing (PDA) layer to balance intrinsic-layer space charge effects and minimize thermal heat loading. Contrasting to traditional devices, these PDA photodiodes (Figure 1.6) generate 10 times higher photocurrents.

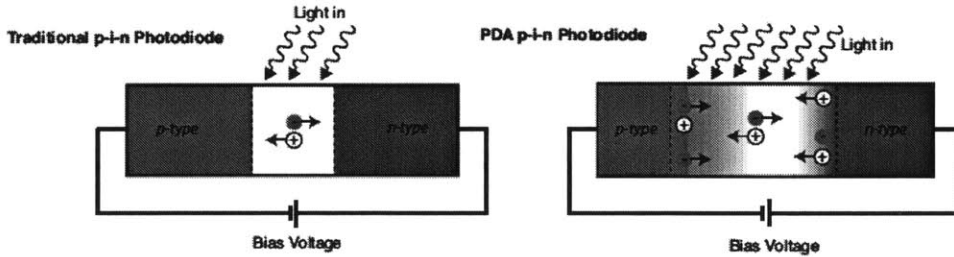


Figure 1.6: Schematic layer structure diagram comparing a $p-i-n$ photodiode to a partially depleted absorber (PDA) photodiode [7].

Newly available photodiodes include photodiodes all made of the same germanium material on a silicon substrate that is transparent to the wavelengths of interest. The consequence of this design is that in addition to

current generated by electron – hole pair drift in the depletion region, photons generated in p and n regions can diffuse into the depletion region and also contribute to current.

Recently, a 42 GHz germanium waveguide has been designed, fabricated and characterized at a wavelength $\lambda = 1.55 \mu\text{m}$ [8]. In this device butt coupling integration has been considered. The rib waveguide width and height are $660 \mu\text{m}$ and $380 \mu\text{m}$, respectively. The dark current of the photodiode is as low as 18 nA at a reverse bias of 1 V. The responsivity at a wavelength of $1.55 \mu\text{m}$ is 0.2 A/W without voltage bias. The quantum efficiency was about 80 %. The measured 3 dB bandwidths were 12 GHz, 28 GHz and 42 GHz at 0, 2 V and 4 V reverse biases, respectively.

We will also mention here the ultra compact 45 GHz CMOS compatible Ge waveguide photodiode, presented by DeRose et al. [9], Figure 1.7. CMOS compatible silicon photonics, as outlined before, has been identified as the most likely candidate for future generation data communication interconnects. A key component here is a photodiode capable of detection of near-infrared light. Photodiode presented in [9] is ultra compact, featuring size of $1.3 \times 4 \mu\text{m}$. It is a Germanium waveguide-based photodiode with best in class 3 dB cutoff frequency of 45 GHz. Due to low capacitance and small device, low dark current is achieved (3 nA). Responsivity of 0.8 A/W confirms the-best-in-class reputation of this device, which may enable “receiverless” optical links with ultra low dissipation in future data communication systems.

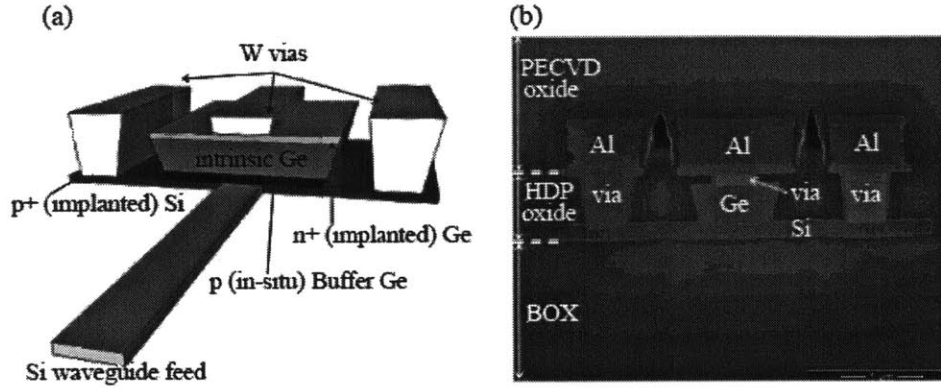


Figure 1.7: (a) Schematic of Germanium waveguide $p-i-n$ photodiode. (b) SEM cross-section of final selective area epitaxially grown Ge photodiode with final electrical contacts [9].

Another important technology of silicone photonics is the one of interconnection techniques. To address high-volume applications with interconnecting the silicone photonic platform, we must develop simple and low-cost coupling and packaging procedures.

In year 2010, Intel researchers have demonstrated their latest breakthrough – silicon-based photonics link running at 50 Gb/s [10], thus bringing Terabit speeds on the horizon. This technology combines fiber-optics with maturity of silicon; unique attributes of laser and IC technologies. This uncovers many advancements in ultra-high bandwidth low-cost optical communications. One can expect for this development to reshape and transform entire computing industry as we know it today.

1.3 Scope of Thesis

Thesis is divided in in the following chapters:

Chapter 2: In Chapter 2 we present the basics of photodetection principles. Absorption in semiconductors is briefly analysed. $P-N$ and $P-i-N$ structures are also analysed. We also provide references for more detailed analysis.

Chapter 3: In Chapter 3 we introduce waveguide topology as very efficient way to couple light from input waveguide into absorbing layer. We analyze two possible configurations for integration of waveguide photodetector. Photodetector characteristics such as efficiency, bandwidth and responsivity have been analysed as well. Also, we discuss advantages of waveguide integrated photodetector and compare it to conventional photodetectors such as vertically illuminated photodetectors.

Chapter 4: In Chapter 4 we analyse various configurations for efficient coupling of light from input waveguide into absorbing layer. Numerical results of performed simulations are in this chapter.

Chapter 5: The last chapter contains the conclusion.

Also, there are two appendices:

Appendix A: We briefly present two simulation techniques used in this

thesis. First we describe the Finite Difference Mode Solver. Then we present a very important technique the Finite Difference Time Domain technique. For each of the simulation techniques we provide references for more details.

Appendix B: This appendix contains two MATLAB codes that illustrates usage of previously described simulation techniques. Section B.1 contains code that uses Mode Solver to find effective index and eigenmode profile of the input waveguide. Code in Section B.2 creates input file for 3D FDTD simulator for simulating light coupling from silicon waveguide to germanium absorption layer. Finally, Section B.3 contains an example of code for use of Meep 3D FDTD simulator.

Chapter 2

Photodetector Basics

A photodetector is optoelectronic device that converts optical signal energy to electrical signal. Photodetector operation is based on photon absorption in semiconductor.

In general, optical signal detection in semiconductor photodetector can be split in three steps:

1. Absorption of optical energy and generation of carriers
2. Transport of generated carriers through absorption layer
3. Photocurrent generation

There are two main types of semiconductor photodetectors:

- Photoconductors, where uniform conductor is used as absorber. Under the illumination electron-hole pairs are generated due to optical excitation. This excess concentration of carriers changes the conductivity

of semiconductor. Under the influence of an external bias voltage optically generated electron-hole pairs are separated and transported in opposite directions. Consequently, photocurrent is flowing.

- Photodiodes, where absorption occurs in depletion region of reversely biased p - n junction. These photodetectors will be further investigated in following sections.

2.1 Absorption in Semiconductors

Absorption involves the interaction of a photon and electron. As the result of the interaction the photon is absorbed and the electron is excited into a higher energy state. When excited, the electrons pass from a bound state to an excited state in which they are mobile. The mobile carriers contribute to current flow.

Absorption in a material is the relative rate of decrease in light intensity, $I(\omega)$, along its propagation direction:

$$\alpha(\omega) = -\frac{1}{I(\omega)} \frac{dI(\omega)}{dx} \quad (2.1)$$

The absorption coefficient for a given photon frequency is proportional to the probability for the transition from the initial state to final state and to the density of the electrons in the initial state and the density of holes in final state. Optical matrix elements for the bulk material can be calculated using Kane's model, which is $\mathbf{k} \cdot \mathbf{p}$ method with spin-orbit interaction taken into account. These matrix elements are used to calculate optical transitions

in semiconductors. Detailed derivation of absorption coefficient using Kane's model can be found in Chapters 4 and 9 of [11]:

$$\alpha(\omega) = \frac{\pi q^2 c \mu_0}{m_0^2 \omega n} |\mathbf{p}_{cv}|^2 \rho_r (\hbar\omega - E_g) \quad (2.2)$$

where q is elementary charge, c is speed of light, μ_0 is vacuum permeability, m_0 is electron mass, n is refractive index, $|\mathbf{p}_{cv}|$ is momentum matrix element for transition from valence band to conduction band, ρ_r is reduced density of states, and E_g is band gap energy.

Absorption coefficient for some of the most important semiconductors is shown in Figure 2.1.

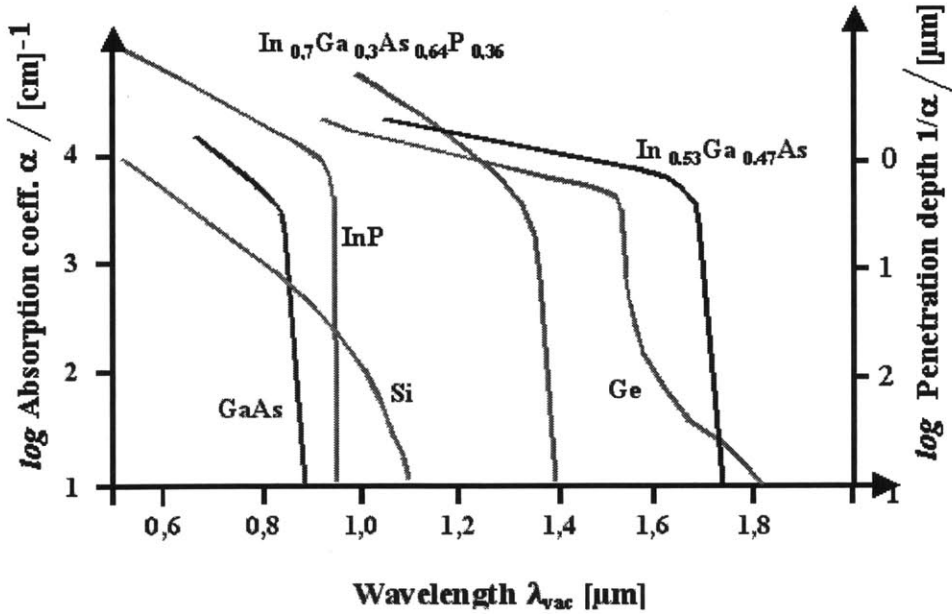


Figure 2.1: Absorption coefficient for some semiconductors [4]

For all optical transitions the requirements of conservation of momentum and energy apply. Therefore transition from valence band to conduction band is only possible when bound carriers interact with a photon whose energy is greater than the band gap of the absorbing material. Also, process of photon absorption conserves the momentum of the excited electron. Consequently indirect transitions are less efficient since they require a two step process involving an optical phonon interaction.

2.2 p - n junction as photodetector

Photodiode is essentially reversely biased p - n junction. In the absence of light, the current through the junction is very low. This current represents the inverse saturation current also known as dark current. When the diode is illuminated by the light of the wavelength equal to the one of the energy gap in the semiconductor from which the photodiode is made of, absorption of the photons occurs along with the generation of electron-hole pairs. The absorption process mostly takes place in the depleted region of the reversely biased pn junction. Due to presence of an electric field, generated electron-hole pairs are separated. In that way, they generate a current which flows from n side of the junction to the p side – photocurrent. This current represents the component of the inverse current through pn junction. Besides photons absorbed in the depletion region, photons absorbed in quasi-neutral (undepleted) regions on distance shorter than diffusion length in from depletion region, also contribute to photocurrent. This happens because such

carriers can also reach depletion region, where they get caught by strong electric field, without being recombined. However, carriers generated in that way slow the photodiode down, because the diffusion velocity is significantly lower than drift velocity and it takes certain time for the carriers to reach space charge region by the diffusion mechanism, before they get caught by electric field. This is why it is needed to broaden depletion region as much as possible in order to increase absorption in this region, and to shorten the quasi-neutral p and n regions in order to minimize absorption there. However, increasing the width of the depletion region will also increase the time carriers need to get through the space charge region, which will also increase response time of photodiode, thus lowering the performance of photodetector.

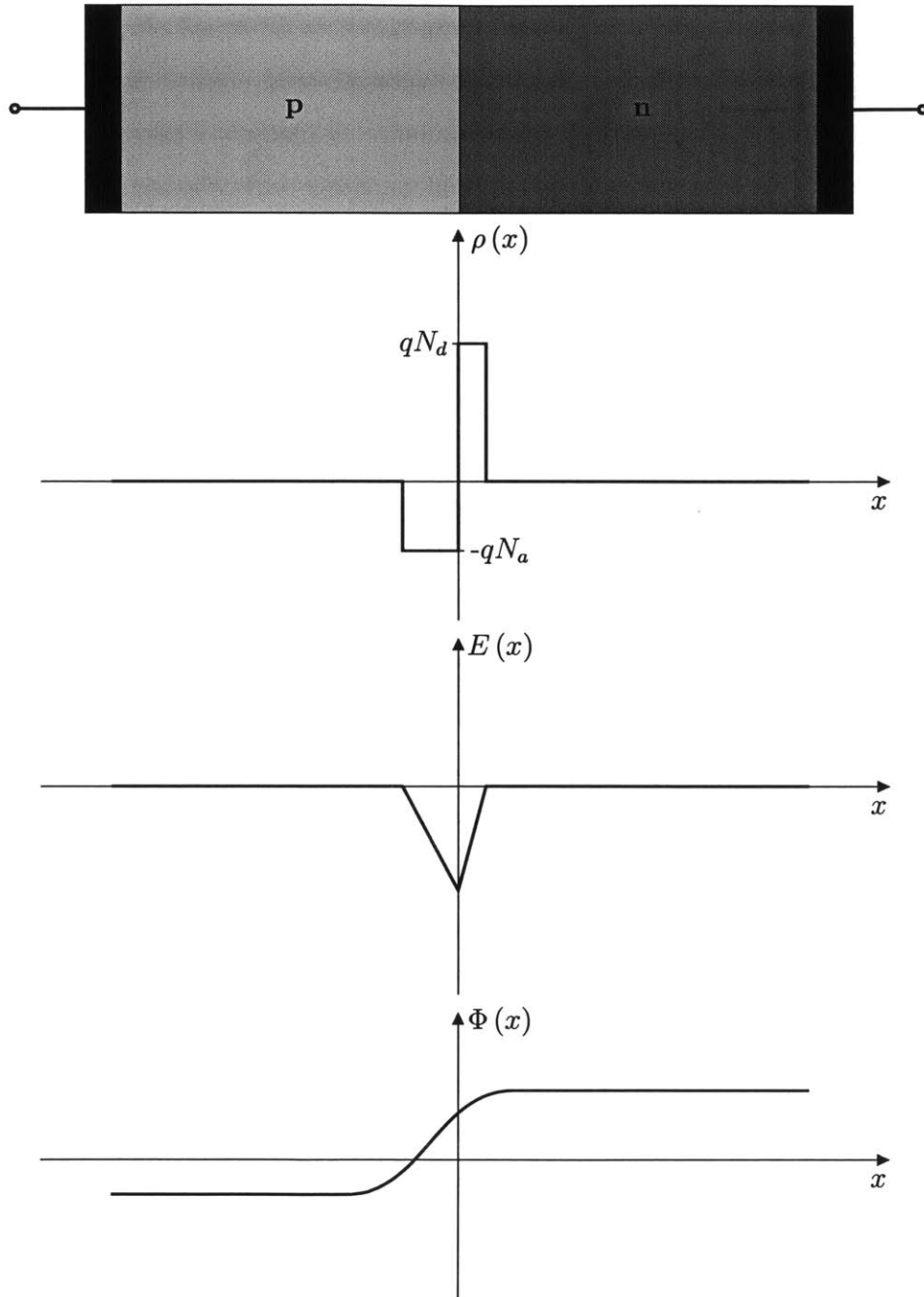


Figure 2.2: Charge density, electric field and potential for a p - n structure

It is common practice to design photodiodes to be based on asymmetrical p - n junction so that the depletion region lies entirely in lightly doped semiconductor. Heavily doped semiconductor is placed to be exposed to the radiation and is usually made of semiconductor with greater band gap, thus preventing absorption in quasi-neutral region. This construction largely eliminates diffusion of the carriers and improves the response.

Figure 2.3 shows p - n junction, uniformly illuminated by photons with energy $E = h\nu$. In the depletion region, with width W , rate of electron-hole pair generation is G_L . Due to strong electric field in space charge region, these pairs are separated: electrons are transferred to n region, while holes are transferred to p region. Photocurrent formed by absorption of photons in space charge region can be found by combining continuity equation with assumptions that the current is purely electron current on left border ($x = x'$) of the space charge region, and purely hole current on the right border ($x = 0$) of the space charge region. We can also assume that, due to a strong electric field, drift is dominant transport mechanism. In such case, transport of the carriers is fast enough to ignore the influence of the recombination of the carriers.

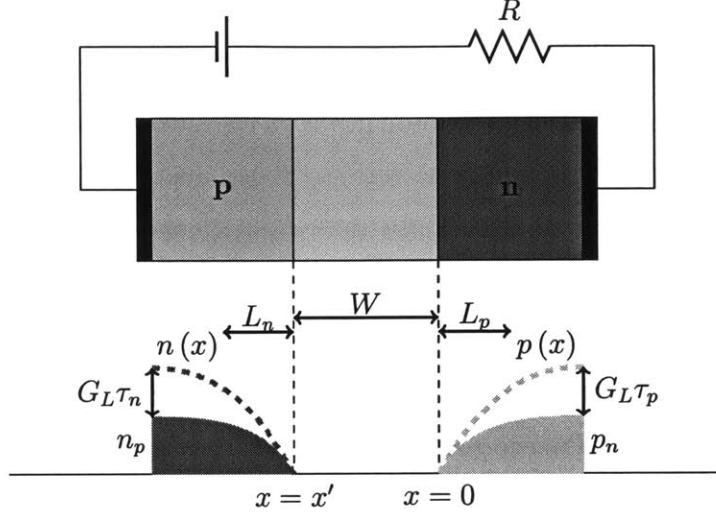


Figure 2.3: Schematic p - n junction and carrier concentrations without illumination (solid) and with illumination (dashed)

If we assume that $G_L(x)$ is determined by uniform distribution of generated electron-hole pairs, photocurrent in depletion region is given by:

$$I_{WL} = qAG_LW. \quad (2.3)$$

Since the movement of electrons and holes which contribute to I_{WL} is governed by strong electric field, response is very fast. This is why this component of the current is often referred to as fast photocurrent.

Beside carriers generated in the depletion area, electron-hole pairs can be generated in quasi-neutral p and n -type regions. One could expect that only holes generated on positions where the distance from space charge region border ($x = 0$) is less than diffusion distance L_p are able to reach space charge region. Then, due to electric field, these are transferred to p side. Simi-

larly, electrons generated in quasi-neutral p -type region on distances shorter than diffusion length L_n from $x' = 0$, are transferred to n side of the pn junction, thus contributing to photocurrent. This means that photocurrent is a consequence of directional movement of all carriers photogenerated in part of the semiconductor with width $W + L_n + L_p$. This can be confirmed through quantitative analysis – by solving continuity equation for given rate of electron-hole pairs generation G_L , one can solve for the expression for photocurrent. Detailed derivation can be found in [12]. Here, we only show the final expression for photocurrent and current through PN junction. The total photo current can be expressed as:

$$I_L = I_{nL} + I_{pL} + I_{WL} \quad (2.4)$$

or

$$I_L = qAG_L (L_p + L_n + W) \quad (2.5)$$

Photocurrent is flowing from n to p side, therefore the total current through p - n junction, when it is illuminated, is given by:

$$I = qA \left(\frac{L_p}{\tau_p} p_n + \frac{L_n}{\tau_n} n_p \right) \left[e^{\frac{qV}{k_B T}} - 1 \right] - qAG_L (L_p + L_n + W) \quad (2.6)$$

or more compactly

$$I = I_s \left[e^{\frac{qV}{k_B T}} - 1 \right] - I_L. \quad (2.7)$$

I - V characteristic of illuminated pn junction for different values of electron-hole pair generation rate G_L is shown in Figure 2.4.

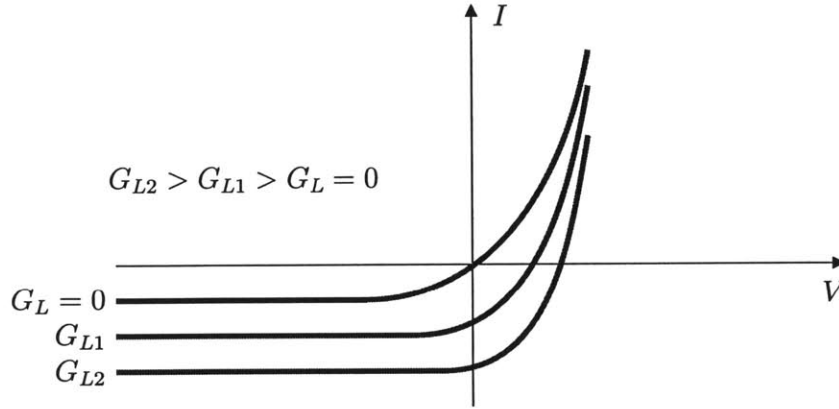


Figure 2.4: I - V characteristic of illuminated p - n junction for different values of electron-hole pairs generation rate G_L

2.3 p - i - n junction as photodetector

p - i - n photodiode, shown in Figure 2.5, is a device similar to standard p - n photodiode. Important difference is that p - i - n photodiode is made of heavily doped p^+ and n^+ -type semiconductor layers with lightly doped or undoped layer in between. Such pn junction will feature very wide depletion region, which makes electric field nearly homogenous in the whole region. In practice, idealized intrinsic region is approximated by a highly-resistive p -type layer (π -layer) or n -type layer (ν -layer). The nature of lightly doped intrinsic region leads to the fact that the greatest voltage drop appears right in this region. Since the p - i - n diode functions in reversely biased regime, electric field in intrinsic region is very strong. This field is controlled by reverse

bias voltage, which is usually chosen to have values just a bit under the diode breakdown voltage.

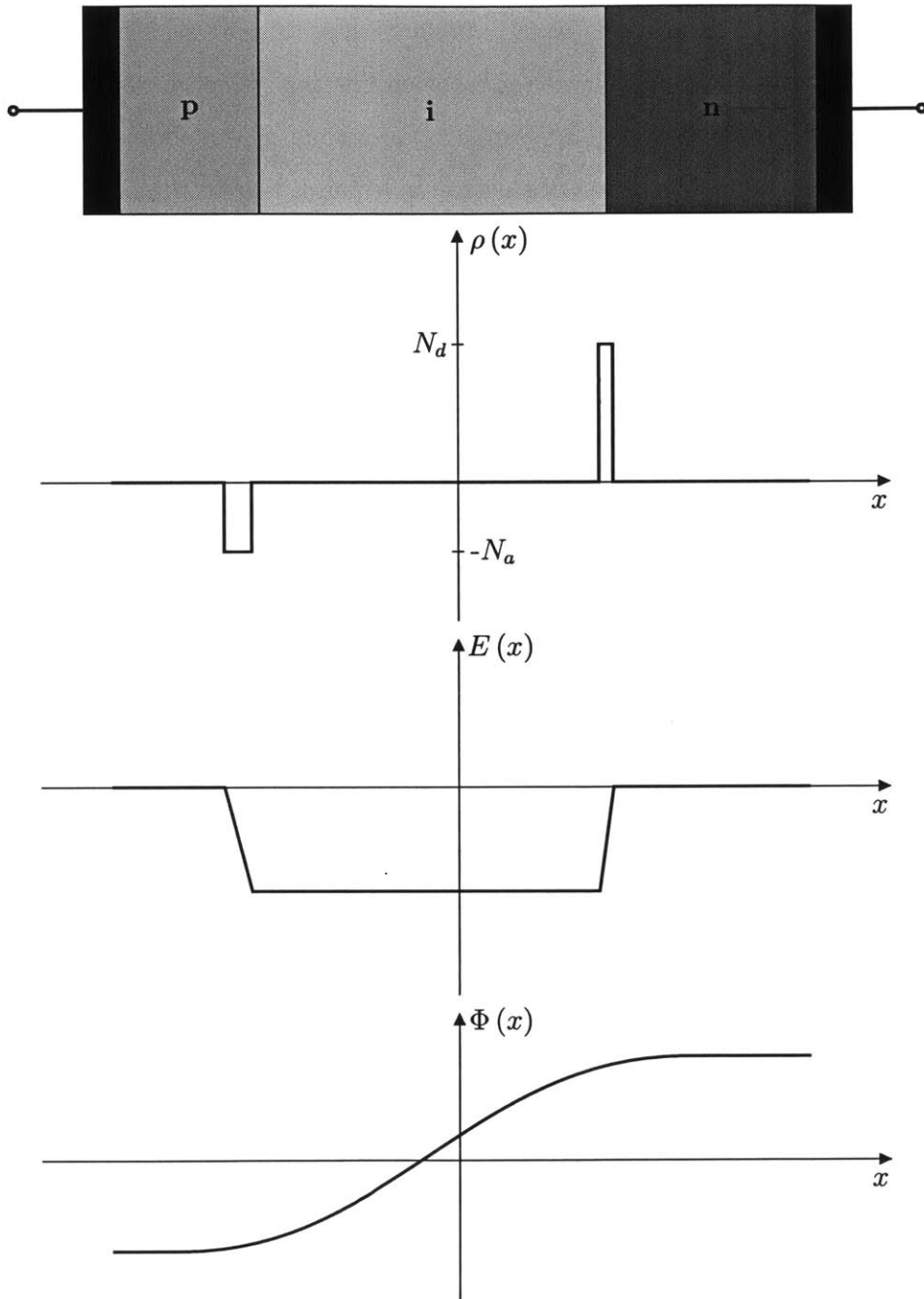


Figure 2.5: Charge density, electric field and potential for a $p-i-n$ structure

The illuminated surface is usually made of very thin p -type semiconductor or it is made of material with greater energy gap, which minimizes the light absorption in this region. Similarly to standard p - n photodetectors, absorption leads to generation of electron-hole pairs. Due to the presence of electric field, these pairs are separated, carriers are taken to opposite electrodes and photocurrent is generated. In this specific case, minor carriers, electrons, are moving towards n^+ region, while holes are moving towards p^+ region. Due to absorption of the light in the semiconductor, the intensity of the light decreases exponentially with distance from the illuminated surface. Hence, the number of generated electron-hole pairs decreases exponentially.

The electric field in the depletion region comes as a result of the fixed charge from p and n dopant atoms on either side of the junction, i.e. $\rho(x) = N_D$ on the n side and $\rho(x) = N_A$ on the p side, assuming uniform doping. The depletion region width and electric field profile depend on doping levels on either side of the junction following Gauss's law:

$$-\frac{d^2\Phi(x)}{dx^2} = \frac{dE(x)}{dx} = \frac{\rho(x)}{\epsilon} \quad (2.8)$$

In the case of the p - i - n junction, in the intrinsic region $\rho(x) = 0$, therefore there is a uniform electric field and most of the change in potential occurs over the i -region. If reverse bias is applied the electric field in the i -region is increased, much like a parallel plate capacitor.

Chapter 3

Modeling Waveguide Photodetector

Most of the photodetectors, including the p-i-n photodetectors, the Schottky photodiodes, the MSM photodetectors and the avalanche photodetectors the optical signal propagates in a direction perpendicular to the junction interfaces of the device. The ultimate limit for the response time for these photodetectors is given by the transit time of the photogenerated electron-hole pairs, it can't be minimised by decreasing the thickness of the depletion region without reducing quantum efficiency (i.e. the fraction of the incident light that is absorbed.) In indirect bandgap semiconductors this problem is more important because the absorption coefficient is smaller than in direct bandgap semiconductors. This geometry leads to a trade-off between the carrier transit time and the quantum efficiency, resulting in a limitation on the bandwidth-efficiency product of the device. Waveguide photodetectors have been developed to overcome this trade-off. In the waveguide photode-

tector light propagates in a direction that is parallel to the junction interfaces and is perpendicular to the drift of the generated electron-hole pairs. This geometry decouples absorption length from the drift length. Therefore the waveguide photodetector can have both a very thin active region for short transit time and a long absorption length for a high quantum efficiency. In integrated photonic circuits photodetector is one of key components. Modern applications require that photodetector has a high 3 dB bandwidth.

3.1 Configuration

There are two typical schemes for the integration of photodetector with waveguide [16]: evanescent coupling and butt coupling. The evanescent coupling scheme is shown in Figure 3.1a. In this case the absorbing material is positioned on top of the waveguide. Incident light couples through the evanescent tails of the waveguide modes. The evanescent coupling scheme provides monolithic process control of the waveguide - detector interface and eliminates the requirement of precise alignment.

In the case of the butt coupling scheme the photodetector is aligned in series with the input waveguide, Figure 3.1b. This scheme leads to a increased photon absorption rate so the required length of the device is shorter. However, precise alignment demands complex fabrication capabilities. Also, reflection at the waveguide - detector interface is not negligible, especially in the high index contrast interface and has to be overcome with an anti reflection coating.

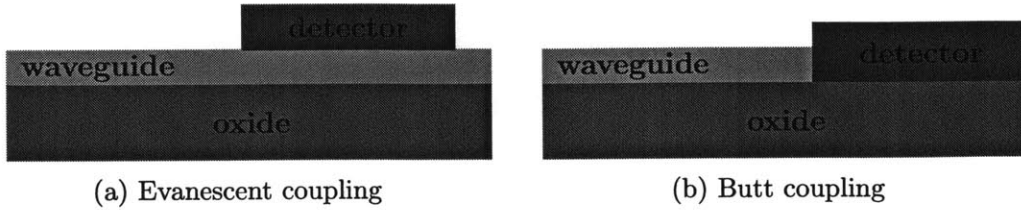


Figure 3.1: Waveguide and detector integration schemes

Side view of waveguide photodetector structure is illustrated in Fig. 3.2

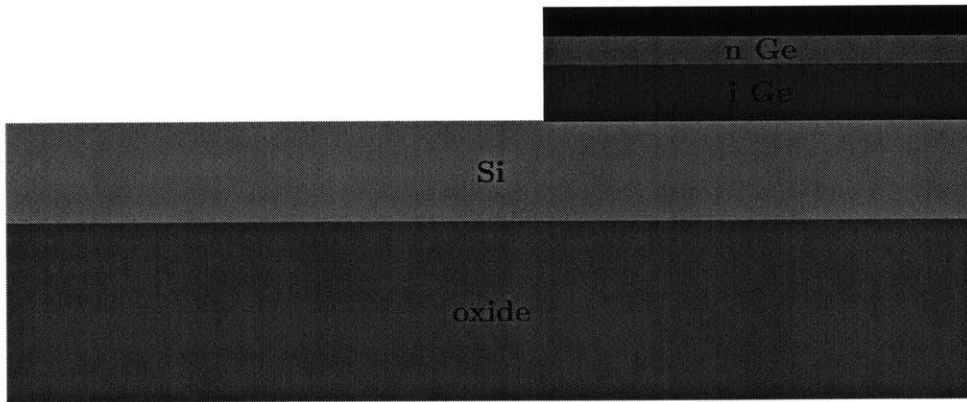


Figure 3.2: Side view of photodetector structure

Three dimensional structure of the detector is shown in Figure 3.3.

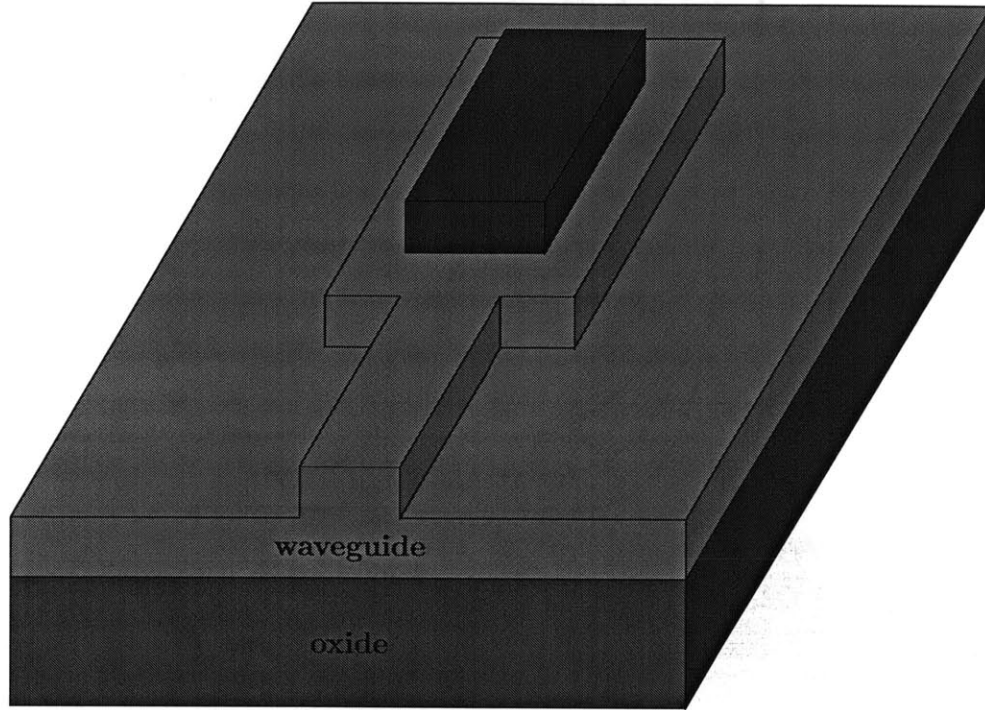


Figure 3.3: 3D model of waveguide photodetector

3.2 Choice of materials

In direct band gap semiconductors (like GaAs, InAs, InP, GaSb, InGaAs) the photon absorption does not require assistance from lattice vibrations. Since the photon momentum is much smaller than electron momentum, the photon is absorbed and the electron is excited directly from valence band to conduction band without change in its k -vector. In indirect band gap semiconductors (Si and Ge) the photon absorption requires assistance from lattice vibrations. Thus the probability of photon absorption is not as high as in direct transition. Absorption coefficient for direct band gap semiconductors

risers sharply with decreasing wavelength, while for indirect semiconductors it is not as sharp, Figure 3.4.

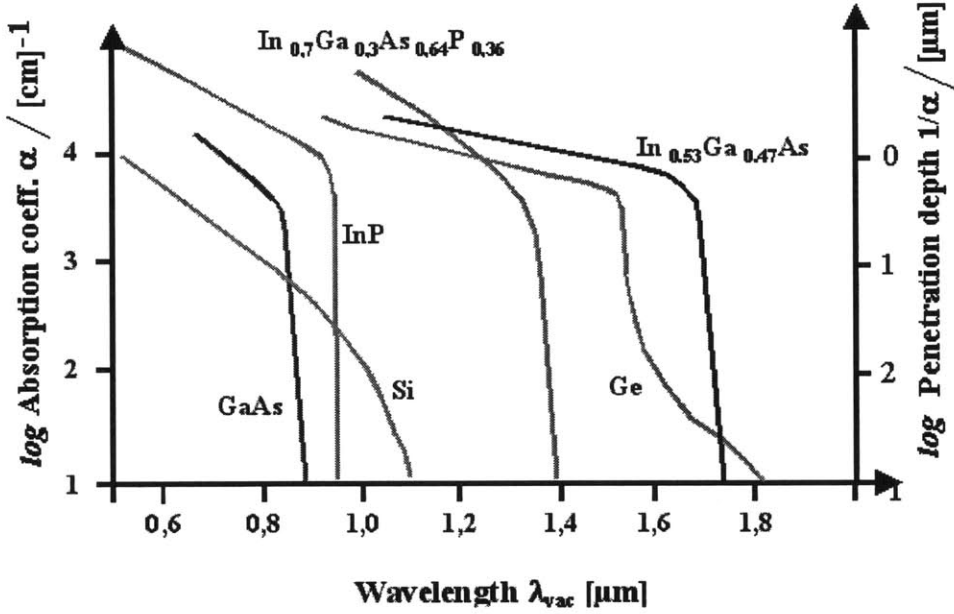


Figure 3.4: Absorption coefficient and penetration depth of various semiconductors as function of wavelength [4]

Although Ge is commonly known as an indirect bandgap material, its direct gap at Γ valley is only 136meV higher than than the indirect bandgap at L valley. The band structure of Ge is shown in Fig. 3.5. The technically most important wavelength in optical communications of 1550 nm corresponds to the direct band gap of 0.8eV. Furthermore, the difference between direct and indirect bandgaps can be reduced by introducing tensile strain. The recent advances in band-engineering by tensile strain enable high performance Ge-on-Si active photonic devices. Majority of current photonic systems operate on wavelengths in the range 1.2 - 1.6 μm . In this range silicon is transparent

and is often used to fabricate waveguides for transmitting optical signals. On the other hand, germanium has very high absorption coefficient over this range and is suitable material for photodiodes. More importantly, germanium is compatible with standard silicon fabrication processes. High absorption at the wavelengths of interest and compatibility with fabrication processes make germanium a very good material choice for making photodetectors at $1.55 \mu\text{m}$.

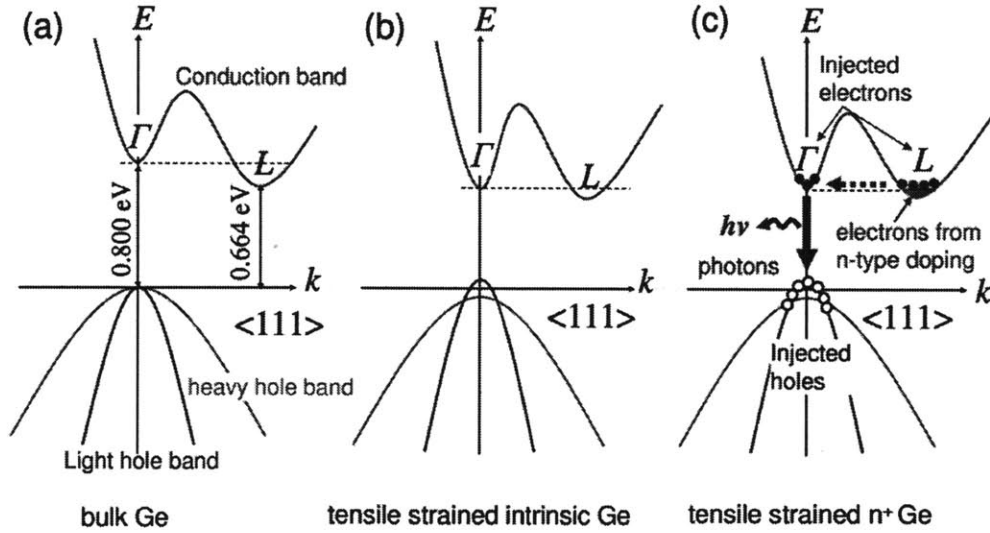


Figure 3.5: Band structure of germanium [13]

Besides all the advantages discussed above, germanium also has several disadvantages which make germanium based integrated devices hard to fabricate. Between pure germanium and pure silicon there is 4 % lattice mismatch that can create significant stress in germanium layers. Another problem when working with germanium is that germanium lacks of a good oxide since GeO_2 is soluble in water [14]. This makes GeO_2 extremely difficult to process.

Furthermore, germanium requires lower processing temperature than silicon does. The melting point of germanium is 937° C compared to 1410° C for silicon [15]. This means that after germanium deposition, processing temperature must be kept low to avoid diffusion of silicon and germanium. Finally, some reactive intermediate compounds of germanium are poisonous.

3.3 Efficiency

Current generated by absorbed photons consist of two major components: diffusion current originating from carriers excited in p and n layers and drift component originated from electron-hole pairs generated in the depletion region. For efficient collection of the generated electron-hole pairs it is necessary that the intrinsic layer is fully depleted.

Because the propagating mode spreads outside the waveguide absorption coefficient is reduced by the mode confinement factor Γ . Therefore, the absorption of the photons contributing to the photocurrent is $\Gamma\alpha$.

The intrinsic quantum efficiency η_i is defined as number of electron-hole pairs created by one incident photon ($\eta_i < 1$) and it mainly depends on the quality of fabrication process. Quantum efficiency η is then

$$\eta = \kappa (1 - R) \eta_i (1 - e^{\alpha_{\text{eff}} L}) \quad (3.1)$$

where κ is the fiber-to-waveguide coupling efficiency, R is the reflection coefficient, L is the length of the photodetector, $\alpha_{\text{eff}} = \Gamma\alpha_{\text{Ge}}$, $\alpha_{\text{Ge}} \approx 4.6 \cdot 10^2 \text{cm}^{-1}$ and

$$\Gamma = \frac{1}{L} \int_0^L \Gamma(z) dz. \quad (3.2)$$

$\Gamma(z)$ is the confinement factor inside active region and it varies along the device, therefore we are using average value. The quantum efficiency has been calculated under the assumption that the fiber-to-detector coupling coefficient is unity.

3.4 Bandwidth

For the purpose of bandwidth calculations we model detector as a parallel plate capacitor. There are two main contributions to photodetector bandwidth. The first one is the RC time constant, the second one is the transit time which is determined by the carrier velocity.

Bandwidth can be estimated using:

$$f_{3dB} = \frac{1}{\sqrt{(R_L C)^2 + \left(\frac{\tau_{tr}}{2.636}\right)^2}} \quad (3.3)$$

where: $R_L = 50\Omega$ is load resistance

$C = \epsilon_0 \epsilon_r \frac{WL}{d}$ - parallel plate capacitor

$\tau_{tr} = \frac{d}{v_{sat}}$ - transit time

$v_{sat} \approx 6 \cdot 10^6 \text{cm/s}$ - saturation velocity

3.5 Equivalent circuit model

Fundamentally a photodiode is a current source. Therefore we can use lumped element circuit abstraction to model it. Equivalent circuit model is very useful to determine frequency characteristics of waveguide photodetectors.

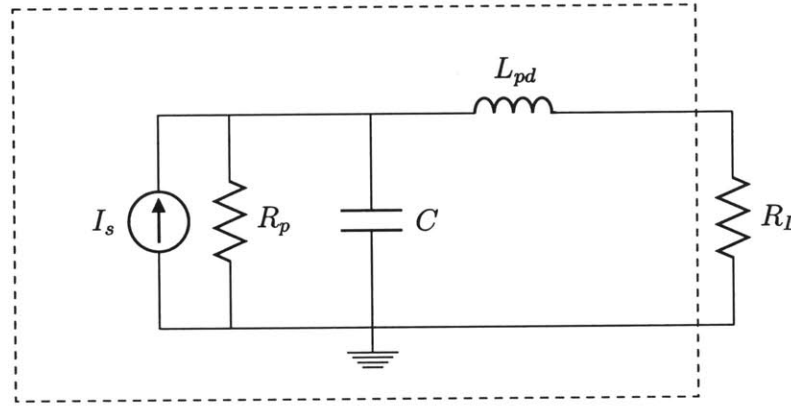


Figure 3.6: Equivalent circuit model

Circuit shown in Figure 3.6 is usually a good model for a waveguide photodetector. The photodiode is modeled as an ideal current source I_s in parallel with the junction capacitance C (calculated in previous section) and resistance R_p . R_L is external load resistor and is usually $50\ \Omega$. In most cases the parallel leakage current is small compared to the generated photocurrent and therefore R_p becomes very high and can be omitted. The inductance L_{pd} may originate from electrical interconnections. It is usually in the pH-range and can be neglected in regimes with bandwidths up to 30 GHz. Standard and well developed techniques for circuit analysis in frequency domain can be applied to our model of the photodiode. This way the influence of parasitic

circuits elements, like series resistance and inductance, on the response speed can be taken into account. These parasitic elements depend on material and fabrication parameters.

3.6 Responsivity

Two main benchmarks for a photodetectors are bandwidth-efficiency product and responsivity. Both of these are directly related to the absorption coefficient and penetration depth of the light. Responsivity is the ratio of collected photocurrent to the optical power incident on the detector. The responsivity, $\mathcal{R}$, of the photodetector can be calculated using the following expression:

$$\mathcal{R} = \frac{q\lambda}{hc}\eta \quad (3.4)$$

where λ is the wavelength, q is the electron charge, h is Planck's constant and c is the speed of light.

3.7 Comparison of vertically illuminated PD and waveguide PD

The following example illustrates the advantage of waveguide photodiode over conventional vertically illuminated photodetector. Both devices are used for the detection of optical signals at $\lambda = 1.55 \mu m$. The intrinsic Ge active region has a thickness of $d = 0.2 \mu m$, and is sandwiched between p doped and n doped semiconductors. The absorption coefficient for Ge at $\lambda =$

$1.55 \mu\text{m}$ is $\alpha = 5.2 \cdot 10^5 \text{ m}^{-1}$. The electron and hole saturation velocities are $v_e = 6 \cdot 10^4 \text{ m/s}$ and $v_h = 5.4 \cdot 10^4 \text{ m/s}$, respectively. The device length is $l = 25 \mu\text{m}$ and width is $w = 2 \mu\text{m}$. With these dimensions, both devices have a capacitance $C = 40 \text{ fF}$, a series resistance from the contacts and the materials of $R_S = 40 \Omega$. The load resistance is $R_L = 50 \Omega$. For the waveguide photodiode, the confinement factor of the active region is $\Gamma = 45\%$.

$$v_{\text{sat}} = \left[\frac{1}{2} \left(\frac{1}{v_e} + \frac{1}{v_h} \right) \right]^{-1} = 5.68 \cdot 10^4 \text{ m/s} \quad (3.5)$$

With $d = 0.2 \mu\text{m}$, we find transient time

$$\tau_{\text{tr}} = \frac{d}{v_{\text{sat}}} = 3.5 \text{ ps}. \quad (3.6)$$

With $C = 40 \text{ fF}$, $R_S = 40 \Omega$ and $R_L = 50 \Omega$ we have

$$\tau_{\text{RC}} = (R_S + R_L) C = 3.6 \text{ ps}. \quad (3.7)$$

Therefore, the 3-dB cutoff frequency

$$f_{\text{3dB}} = \frac{0.443}{\sqrt{\tau_{\text{tr}}^2 + \tau_{\text{RC}}^2}} = 88.2 \text{ GHz}. \quad (3.8)$$

Quantum efficiency for the single-pass vertically illuminated photodetector is

$$\eta = 1 - e^{-\alpha \cdot d} = 9.8 \% \quad (3.9)$$

Therefore, its bandwidth-efficiency product is

$$\eta \cdot f_{3\text{dB}} = 8.71 \text{ GHz.} \quad (3.10)$$

For the double-pass vertically illuminated photodetector we have

$$\eta = 1 - e^{-2\alpha \cdot d} = 18.8 \% \quad (3.11)$$

and its bandwidth-efficiency product is

$$\eta \cdot f_{3\text{dB}} = 16.56 \text{ GHz.} \quad (3.12)$$

For the waveguide photodiode, we find that $\alpha_{\text{eff}} = \Gamma\alpha = 3.38 \cdot 10^5 \text{ m}^{-1}$. Quantum efficiency is

$$\eta = 1 - e^{-\alpha_{\text{eff}} \cdot l} = 97 \% \quad (3.13)$$

Therefore, the bandwidth-efficiency product of the waveguide photodiode is

$$\eta \cdot f_{3\text{dB}} = 82.3 \text{ GHz.} \quad (3.14)$$

We find that the bandwidth-efficiency product of the waveguide photodiode is 4.6 times that of the double-pass vertically illuminated photodiode and is more than eight times that of the single-pass vertically illuminated photodiode though all of them have the same 3-dB cutoff frequency.

Chapter 4

Results

4.1 Input waveguide

The incident light on photodetector comes from the waveguide where it is confined in transverse plane and has some mode profile. In our device we use a silicon rib waveguide as the input waveguide. Cross section and fundamental TE mode profile of the input waveguide are shown in Figure 4.1. It is desirable that waveguide is single mode, however it is not necessarily as long as light gets absorbed in intrinsic region.

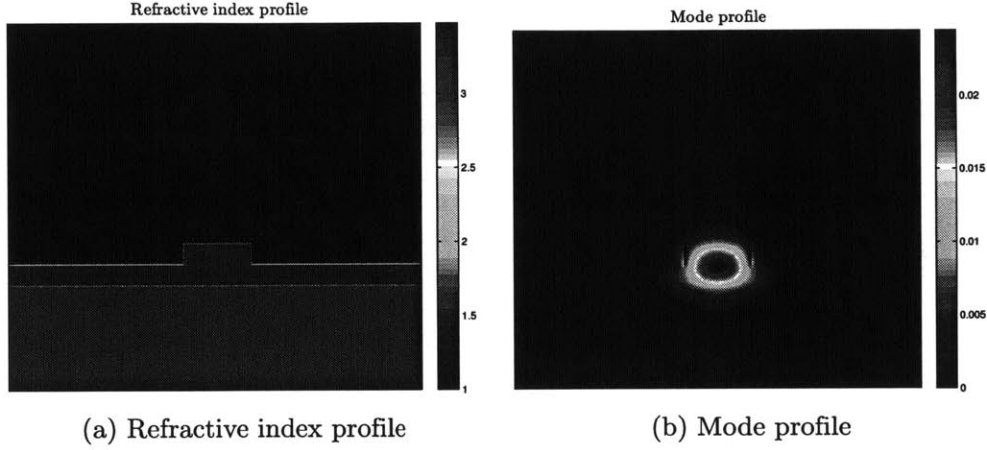


Figure 4.1: Cross-sectional refractive index profile and fundamental TE mode profile of the input waveguide

Flux Monitors 1 and 2 shown in Figure 4.2 are used to calculate power near the input and output side.

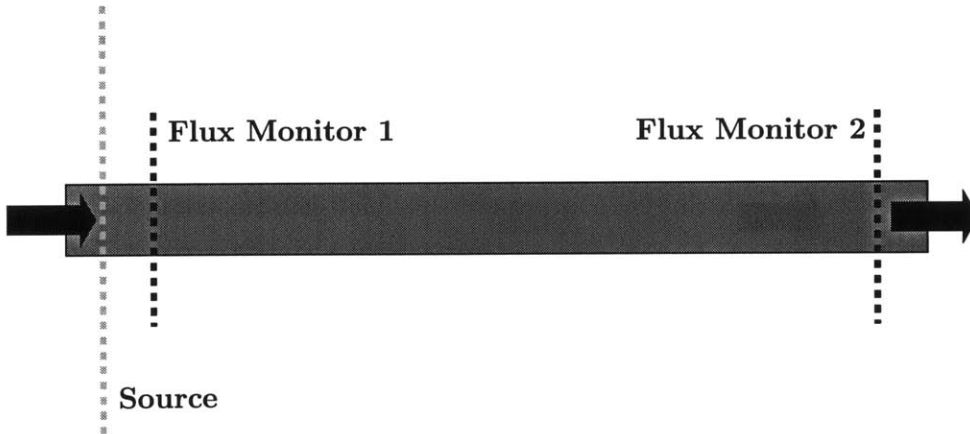


Figure 4.2: Top view of input waveguide in FDTD simulation. Width of the waveguide is $1\ \mu\text{m}$ and length is $16\ \mu\text{m}$.

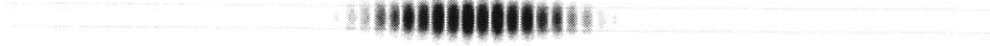


Figure 4.3: Snapshot of wave propagating in the waveguide (top view)

Transmission spectrum at the end of the waveguide (Flux Monitor 2) is shown in Figure 4.4

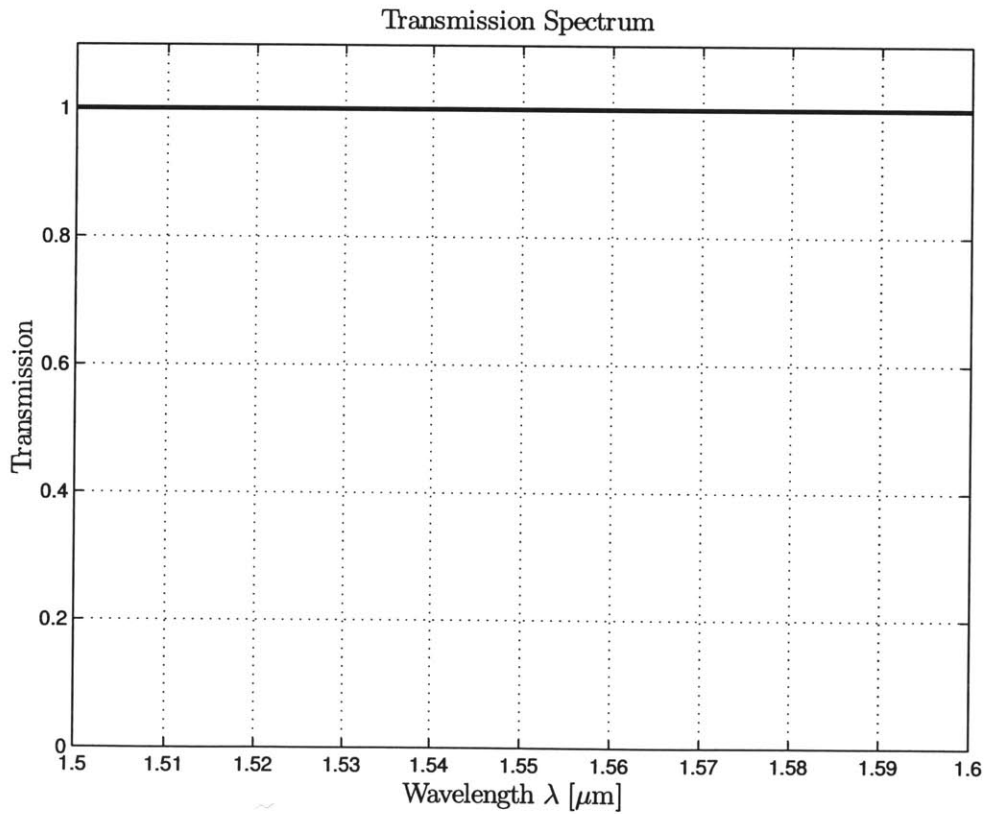


Figure 4.4: Transmission spectrum of the input waveguide

Since we launched an eigenmode it is not surprising that the transmission is 100 % in a narrow bandwidth around $\lambda = 1.55 \mu\text{m}$.

The waveguide height is 400 nm and width is $1 \mu\text{m}$. Two versions of evanescent coupling configurations have been analysed to couple the input light to the Germanium absorption layer. 3D FDTD simulations have been performed to evaluate the efficiency of both coupling configurations.

The electric field amplitude calculated at the wavelength $\lambda = 1.55 \mu\text{m}$ in a longitudinal cross section of the device is shown in Figure 4.5. Because the refractive index of germanium is higher than refractive index of silicon at wavelength $\lambda = 1.55 \mu\text{m}$, the optical mode of the input silicon waveguide is progressively coupled towards the germanium layer.

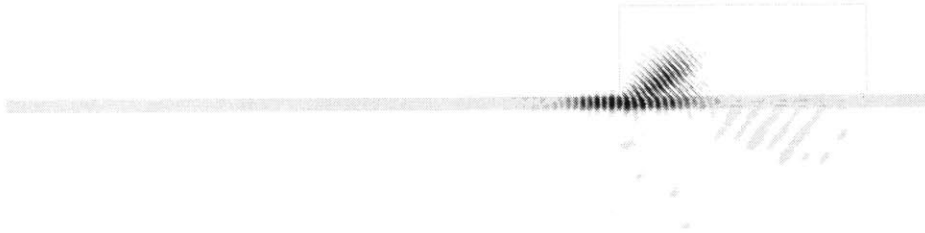


Figure 4.5: Snapshot of coupling from silicon waveguide to germanium layer (side view)

Germanium is grown on top of a silicon rib waveguide by epitaxial process. A cross section schematic of the photodetector is shown in Figure 4.6. As

the light propagates in the Silicon waveguide via total internal reflection it evanescently couples into the Germanium region where it is absorbed.

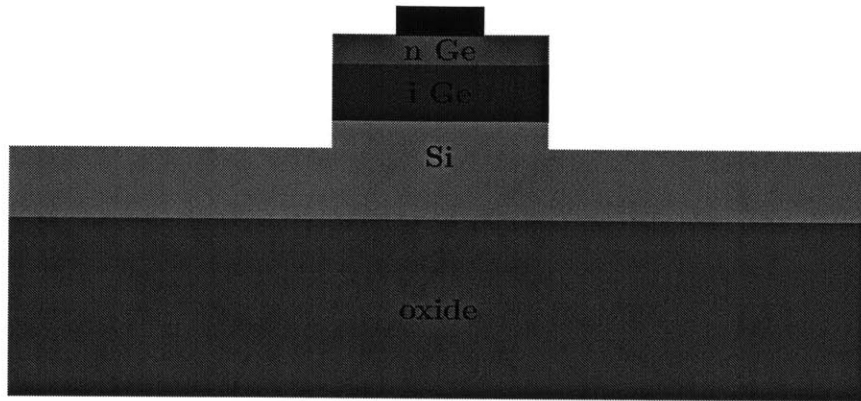


Figure 4.6: Cross-section schematic of the photodetector

Step configuration. Ahn et al [17] found that an offsetting step in the waveguide at the transition interface from the input waveguide to absorbing layer can improve coupling. Coupling in this configuration is therefore more efficient and length of the detector is shorter. Schematic of this coupling configuration is shown in Figure 4.7. The main advantage of this configuration is shorter absorption length.

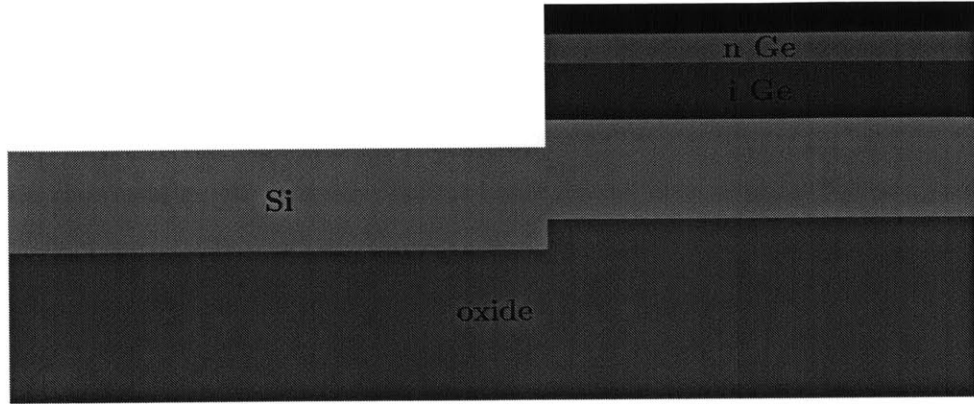


Figure 4.7: Side view of photodetector structure with step abrupt

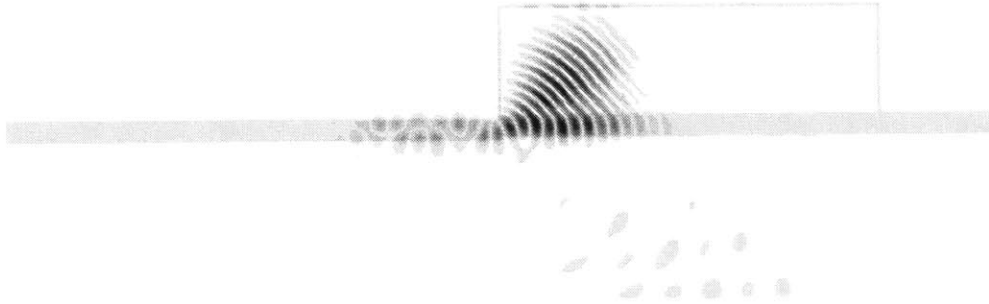


Figure 4.8: Snapshot of coupling from silicon waveguide to germanium layer with step (side view)

In this case coupling is more efficient. However, the reflection in this coupling configuration is not negligible. Even with small step height of 5 % of the waveguide height reflection is significant. Although the variation in waveguide dimension is small, the variation of refractive index is almost 60 %. Therefore, introducing step in the waveguide structure can not be treated as

small perturbation. Consequently, the reflection at the transition interface is hard to predict. Reflection spectrum for three values of step height h is shown in Figure 4.9.

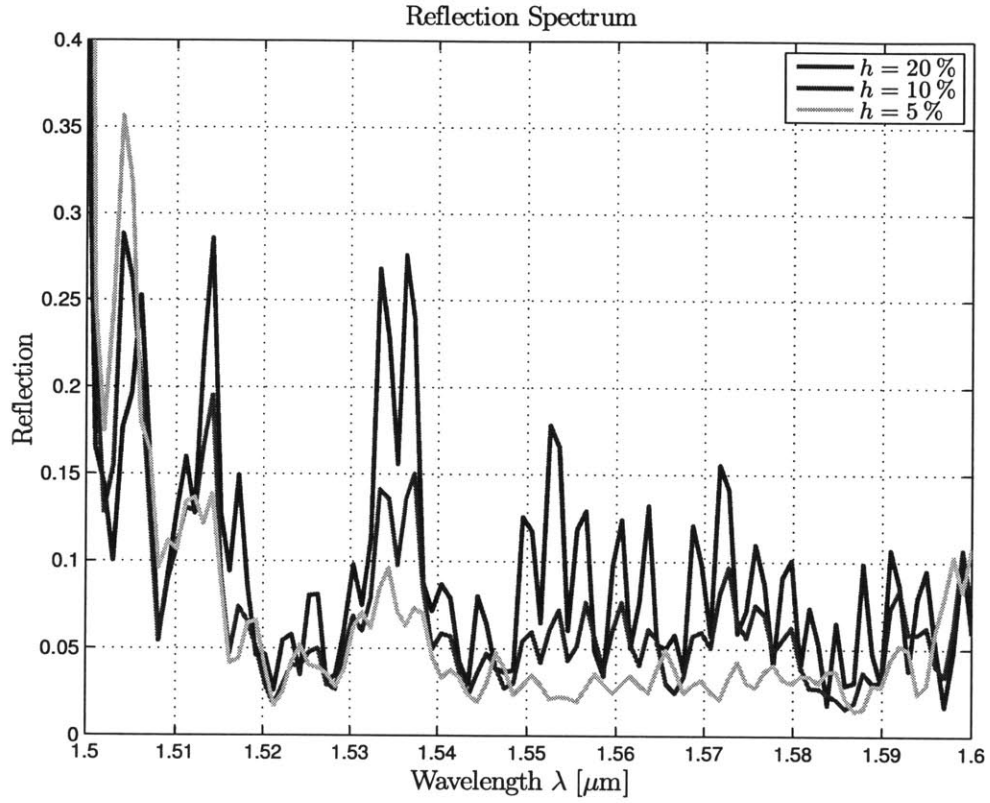


Figure 4.9: Reflection Spectrum for three values of step height h (expressed as percentage of silicon waveguide height)

4.2 Efficiency

Figure 4.10 shows quantum efficiency vs. length of the device.

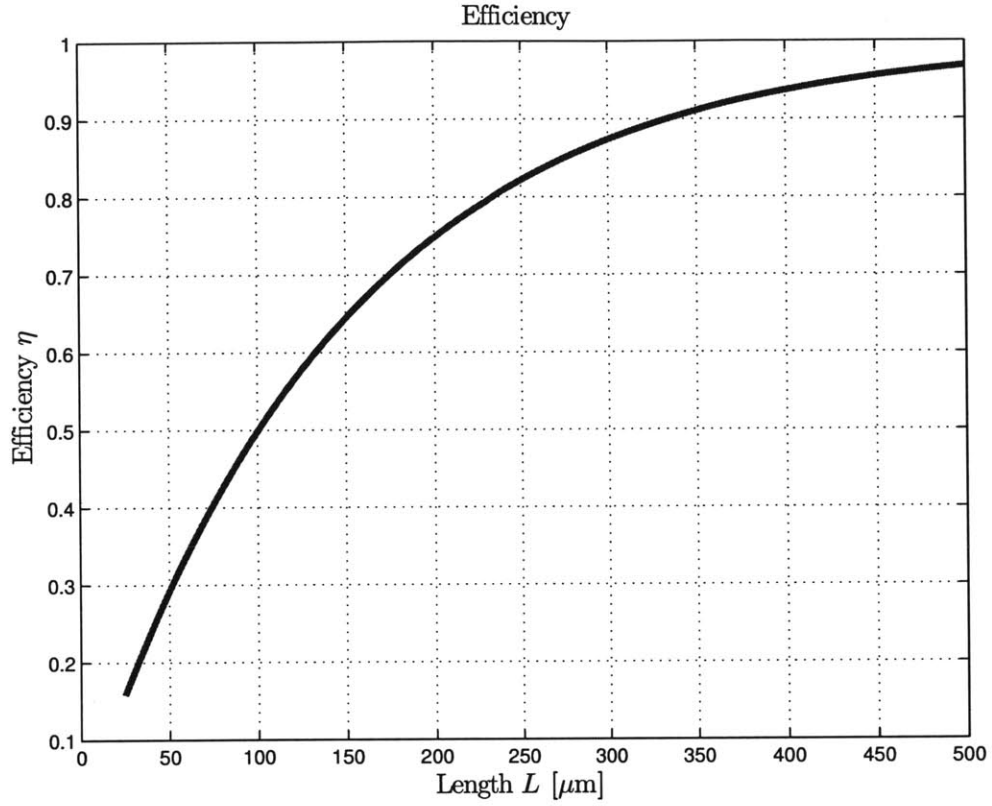


Figure 4.10: Efficiency vs. waveguide photodetector length L

4.3 Bandwidth

Using equation 3.3 we can plot 3 dB bandwidth as function of intrinsic layer thickness d . The variation of 3 dB bandwidth of waveguide photodetector with absorption layer thickness d for different device lengths L is shown in Figure 4.11.

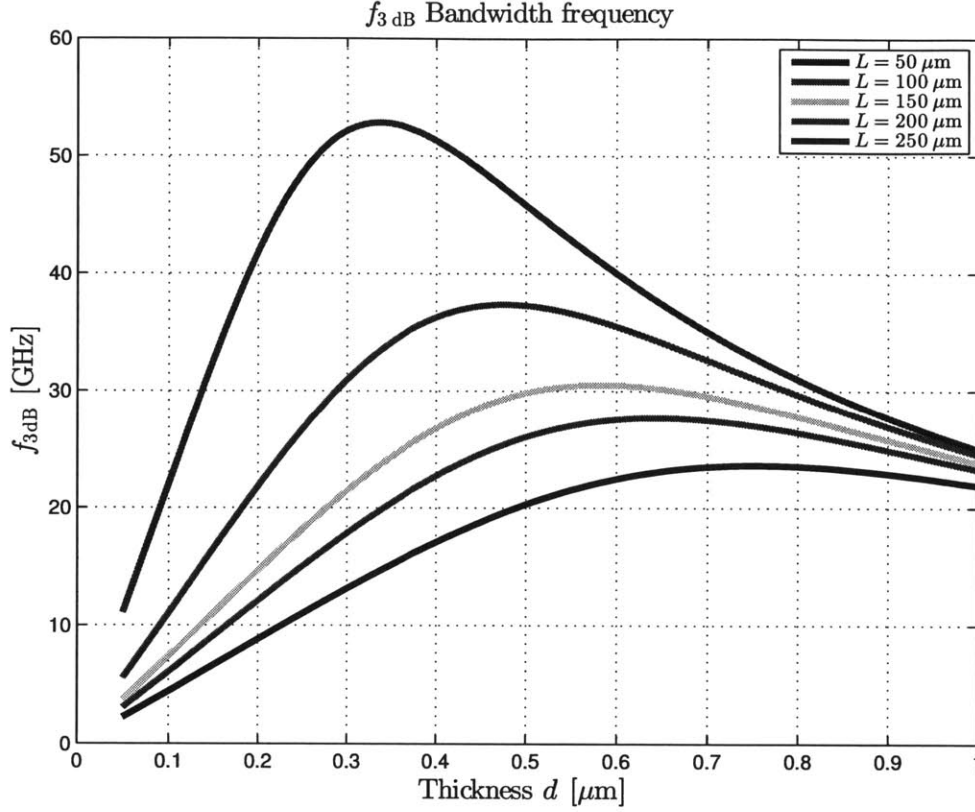


Figure 4.11: Bandwidth vs. thickness for different device lengths

These curves show an optimum design thickness for each device length. If the absorption layer is too thick, transit time effects will be dominant and they will limit the speed. If the absorption layer is too thin, the RC term will be dominant and bandwidth will be limited by the junction capacitance. It seems that, in principle, bandwidths of more than 50 GHz can be easily achieved with shorter device lengths. However, device with shorter lengths would have lower efficiency, Figure 4.10. Therefore, we should use bandwidth-efficiency product as the more reliable figure of merit.

Bandwidth-efficiency product as a function of thickness for various lengths is shown in Figure 4.12.

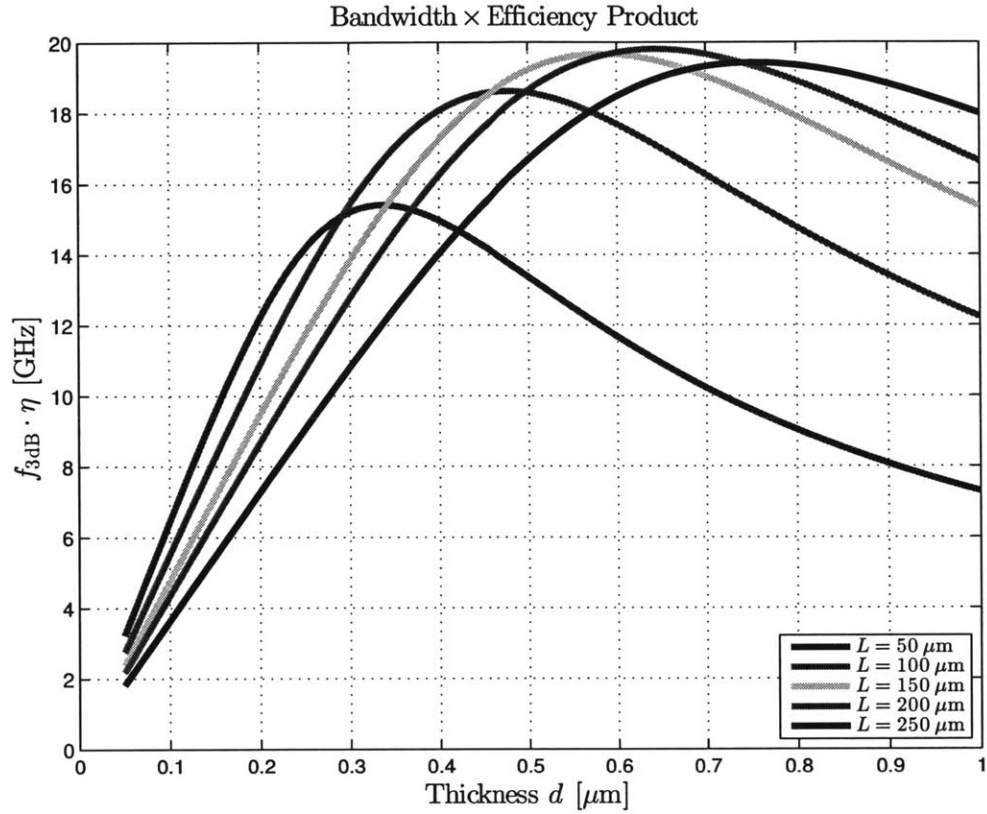


Figure 4.12: Variation of bandwidth-efficiency product with intrinsic layer thickness for different device lengths L

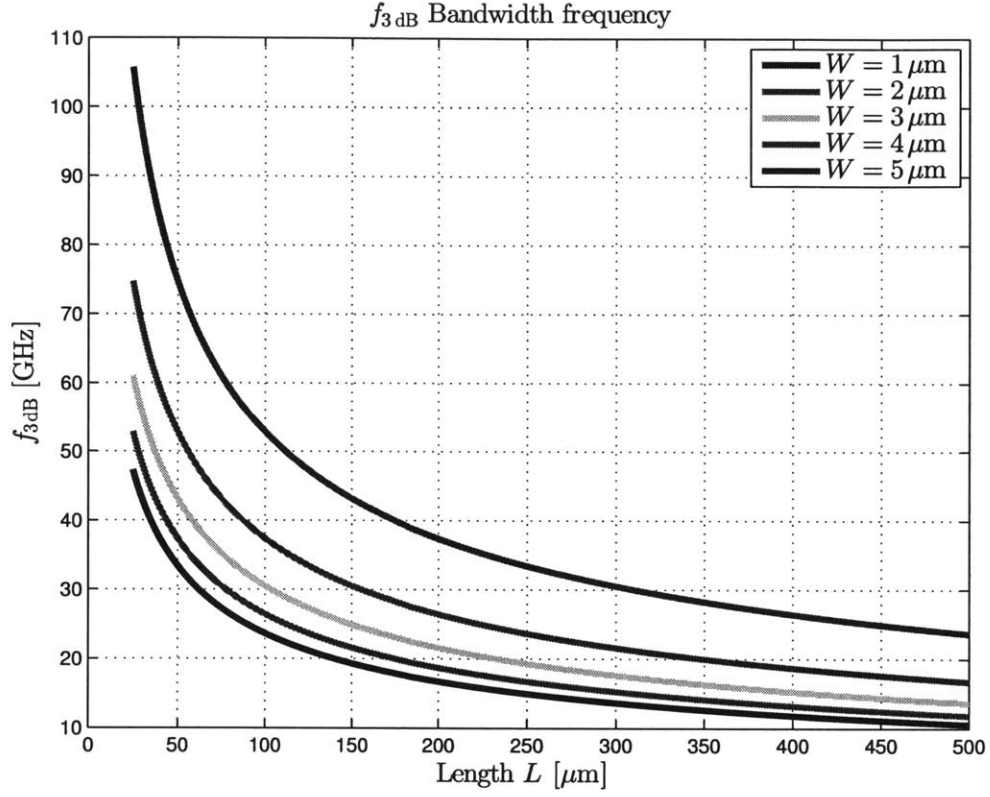
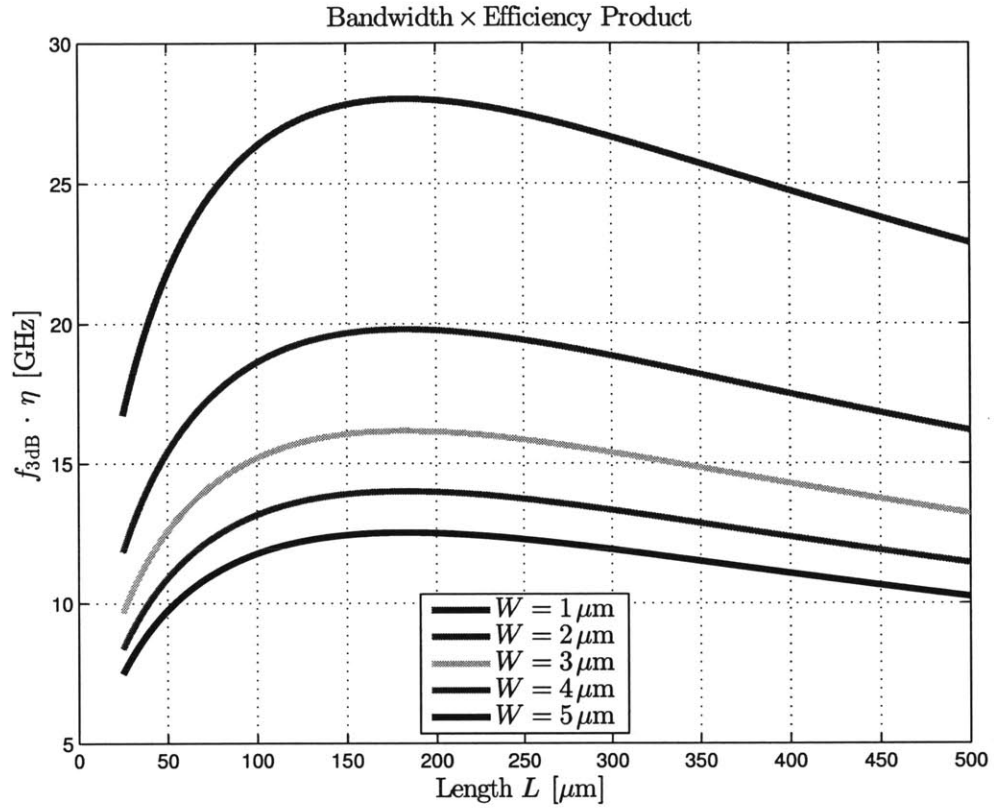


Figure 4.13: Bandwidth vs. length for different device widths W

Bandwidth-efficiency product as a function of length for various widths is shown in Figure 4.14.

Figure 4.14: Bandwidth-efficiency vs. length for different device widths W

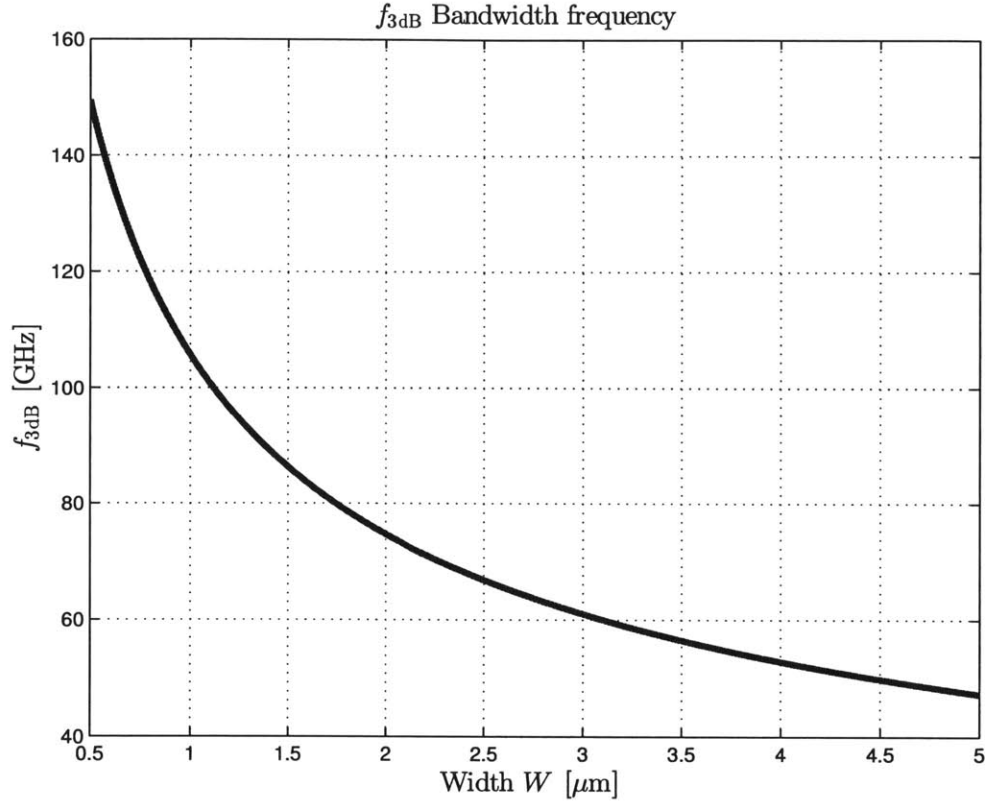


Figure 4.15: Bandwidth vs. width W of the device

Finally, the bandwidth-efficiency vs. waveguide width is shown in Figure 4.16.

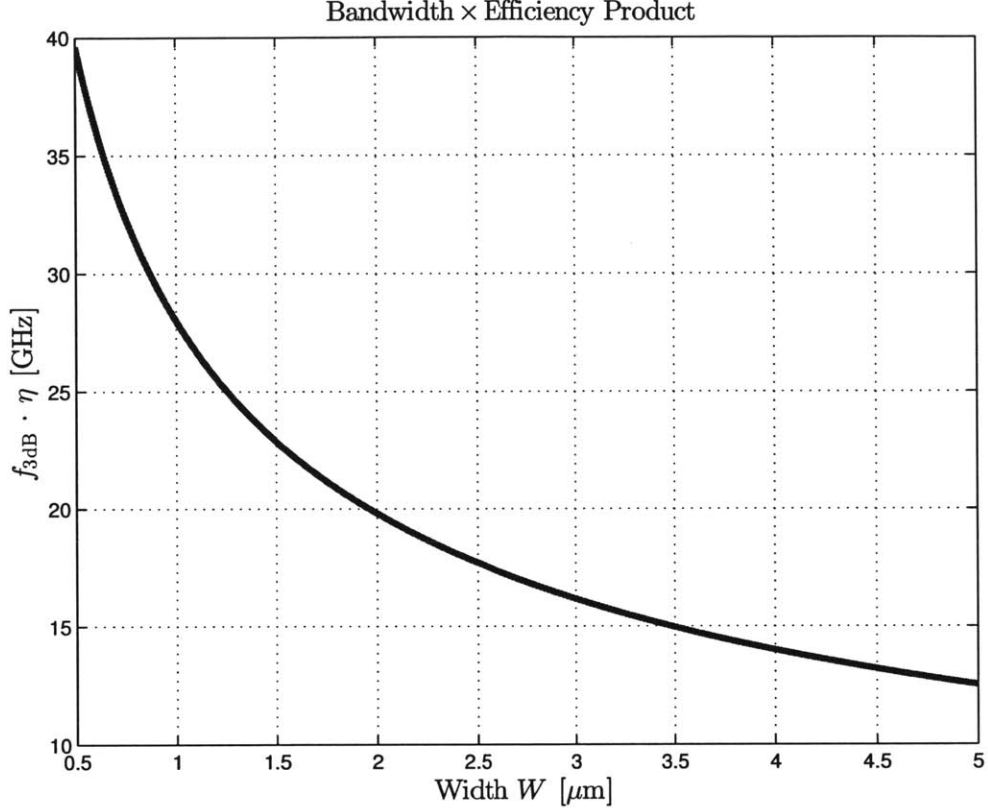


Figure 4.16: Bandwidth-efficiency vs. device width

Note that in this section we have used fixed value for confinement factor of $\Gamma = 15\%$. This is rather low value for confinement factor so the estimated bandwidth $\times$ efficiency product is our lower bound. Confinement factor Γ depends on the geometry of the device and optimal dimensions depend on confinement factor. Therefore, we have to solve for optimal dimensions of the device self consistently. We use optimal device dimensions that we have found for our initial value of $\Gamma = 15\%$ to calculate new confinement factor Γ , then calculate new optimal values for device dimensions, find new Γ , etc. This

process is repeated until the error is less than 10 % i.e. until the difference between new and old Γ is less than 10 %.

After the iteration process is over we find that confinement factor is $\Gamma = 56\%$. With this value of Γ we re-calculate optimal device parameters. New value of Γ only affects calculations that involve efficiency. Therefore, we are plotting again only bandwidth $\times$ efficiency product versus device dimensions.

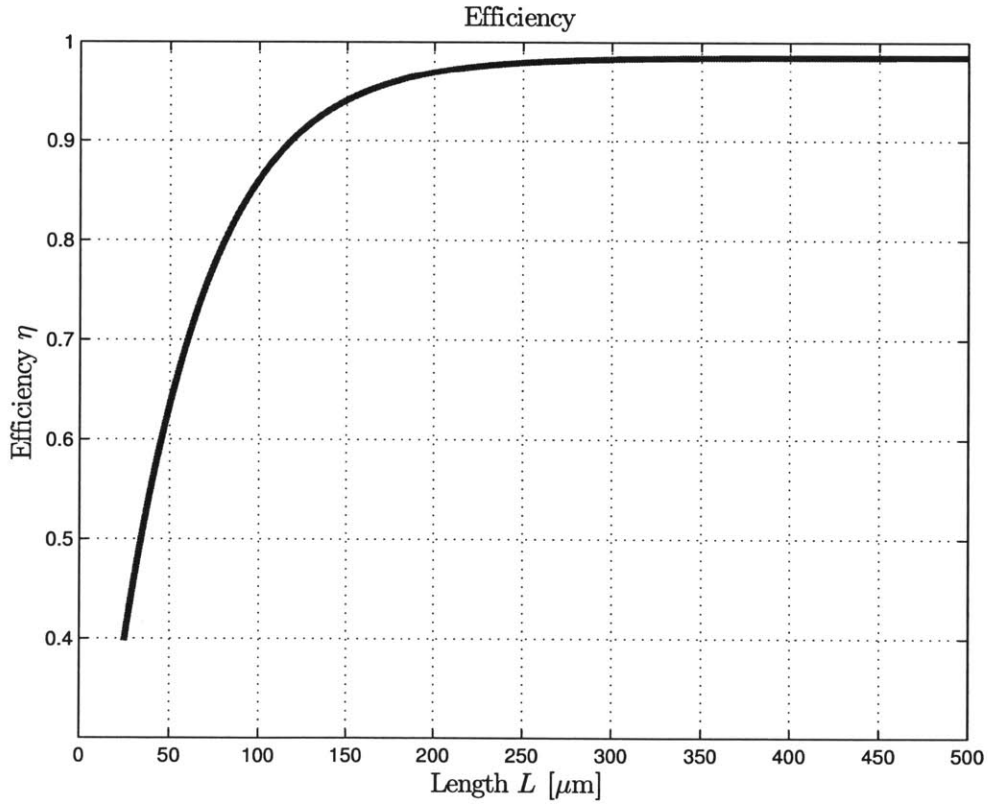


Figure 4.17: Efficiency vs. device length for $\Gamma = 56\%$

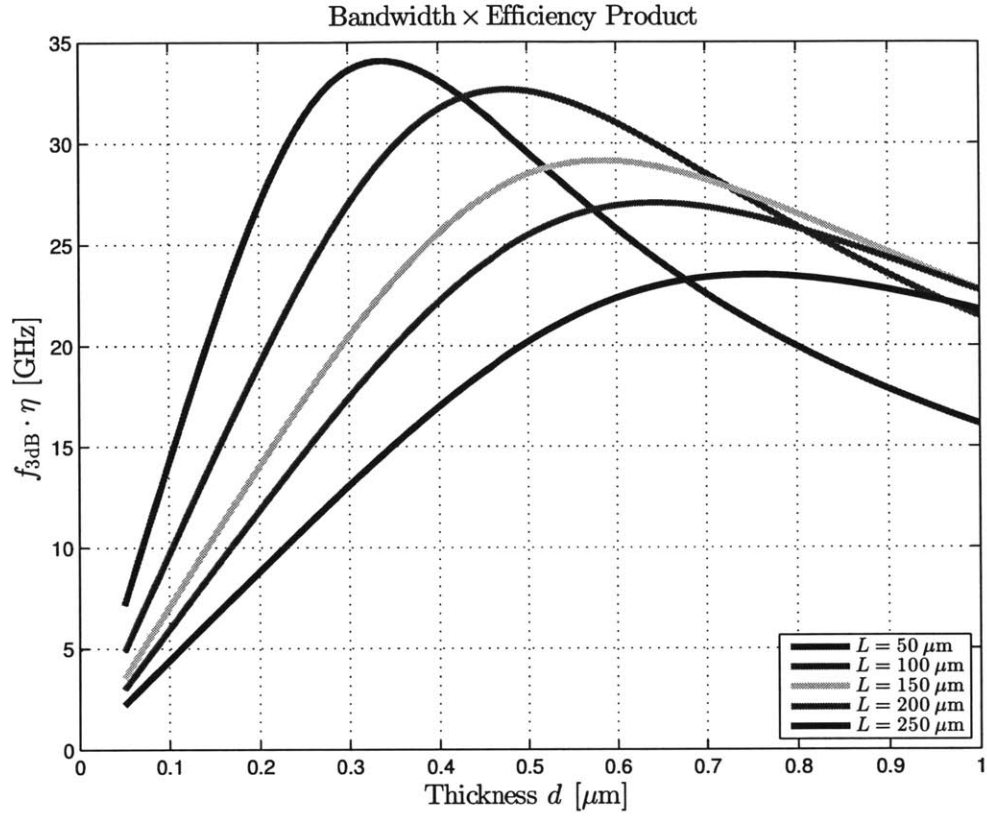


Figure 4.18: Bandwidth-efficiency vs. device thickness for $\Gamma = 56\%$

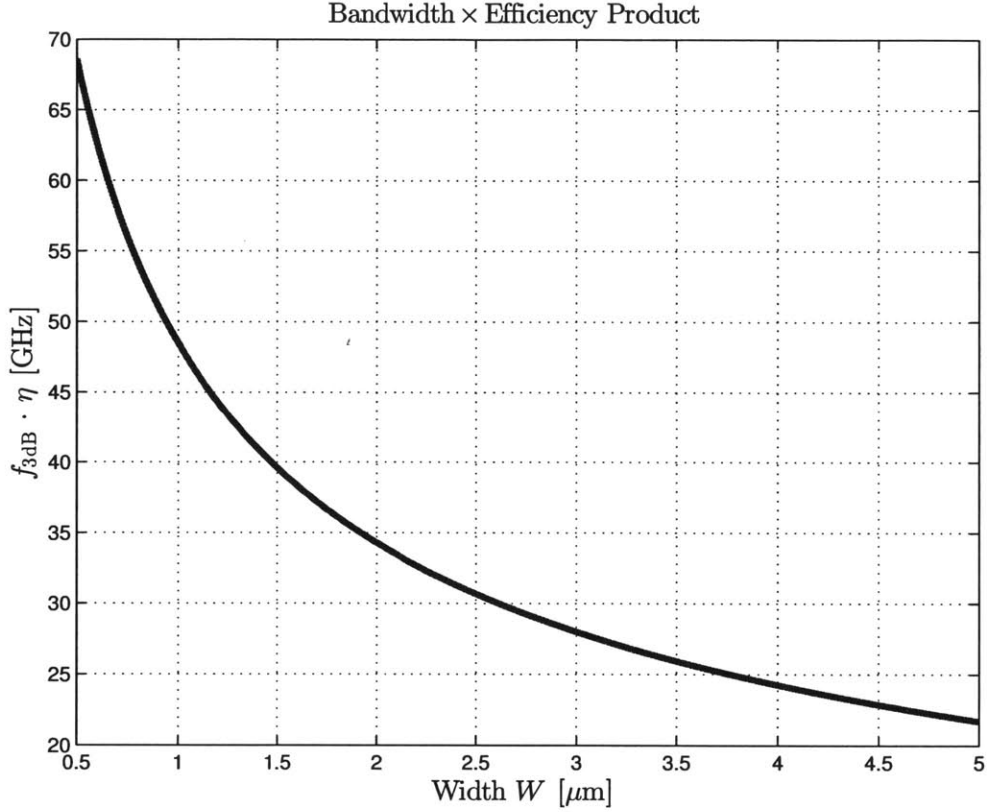


Figure 4.19: Bandwidth-efficiency vs. device width for $\Gamma = 56\%$

Now we can choose width W of the device. Too narrow waveguide will not be able to carry to much power. On the other hand, too wide and the waveguide will operate in multi mode regime. We pick width W to be $1\mu\text{m}$ which is just below cut off so that the waveguide will operate in single mode regime. For this width of the waveguide bandwidth $\times$ efficiency product is 48.5GHz . Now we look for optimal length and find that optimal length of the device is $L = 61.6\mu\text{m}$. Efficiency that corresponds to this length is approximately 77% . Finally, we find that optimal thickness of the absorbing

layer is $d = 370$ nm.

In this section we used bandwidth $\times$ efficiency as the figure of merit to optimise the device parameters. However, the ideal device dimensions can be also defined as those that will maximize bandwidth $\times$ responsivity product. Instead of optimising bandwidth $\times$ efficiency product, we could optimize bandwidth $\times$ responsivity product, analysis similar to analysis we have done in this chapter can be performed in that case. However, that analysis is out of the scope of this thesis.

Chapter 5

Conclusion

In this thesis, we have investigated optimal designs of waveguide integrated photodetectors. We have explicitly showed the advantage of waveguide photodetector over conventional vertically illuminated photodiode in Chapter 3.

In Chapter 4 we have analyzed the limiting factors on both bandwidth and quantum efficiency. We have investigated the influence of waveguide photodetector dimensions on efficiency, bandwidth and bandwidth-efficiency product. Methods of design for different coupling configurations were described in Chapter 4. Also, we have investigated effects of an abrupt step in the waveguide.

Understanding of the waveguide based photodetectors is crucial for making energy efficient photonic circuits with high bandwidth. Because this performance can be achieved without applying a reverse bias voltage, waveguide integrated photodetectors are very well suited for power efficient integrated photonic circuits.

Over the past several years impressive results have been obtained on germanium waveguide integrated photodetectors. The technological processes used to fabricate such devices is completely compatible with technologies developed for integrated microelectronic circuits. Future developments will focus on specific requirements for applications such as low cost optical communications or very high speed optical links between cores of microprocessors. Successful integration of germanium with silicon is the key for future development of silicon photonics. Silicon photonics is an emerging technology offering the potential for orders of magnitude improvements in bandwidth and power efficiency. The final goal is to have silicon photonics participating in every application of the photonics industry including computing, communications, optical signal processing, information displays, optical storage and many others.

Appendix A

Simulation Techniques

In this thesis, two simulation techniques, Finite-Difference Modesolver (FDMS) and Finite-Difference Time-Domain (FDTD) were used to design optical components. Here, we present each of these techniques and provide references for more detailed explanations.

A.1 Finite-Difference Modesolver

Finite-Difference modesolvers are especially convenient as they are easy to implement, reasonably fast and very stable. Here, we provide just a basic overview of the method. The eigenmodes of a waveguide are the solutions to the vector wave equation for the waveguide geometry:

$$\nabla \times \nabla \times \mathbf{E} - \omega^2 \mu \epsilon \mathbf{E} = \nabla (\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} - \omega^2 \mu \epsilon \mathbf{E} = 0 \quad (\text{A.1})$$

Generally, the solutions consist of the eigenvalue, namely the propagation constant β_m and the field distributions $\mathbf{E}_m$ and $\mathbf{H}_m$ where m denotes the

mode number. The transverse electric field components are sufficient to fully define the field solution since the longitudinal electric field component can be derived from Gauss' Law ($\nabla \cdot \epsilon \mathbf{E} = 0$). The transverse wave equation can be extracted from A.1 and written with only the transverse components by using Gauss' Law to express the longitudinal component in terms of the transverse field.

$$\frac{\partial}{\partial z} E_z = -\frac{1}{\epsilon} \nabla_T \cdot \mathbf{E} \quad (\text{A.2})$$

and inserting into A.1 we arrive at

$$\nabla_T \left(\nabla_T \cdot \mathbf{E}_T - \frac{1}{\epsilon} \nabla_T \cdot \epsilon \mathbf{E}_T \right) - \nabla_T^2 \mathbf{E}_T - \omega^2 \mu \epsilon \mathbf{E}_T = -\beta^2 \mathbf{E}_T \quad (\text{A.3})$$

Since analytic solutions to A.3 do not generally exist, A.3 is discretized using central differences across the computational domain. The result is a matrix eigenvalue problem of the form $\mathbf{M} \mathbf{E}_T = \beta^2 \mathbf{E}_T$ where β^2 is the eigenvalue and $\mathbf{M}$ is the matrix operator applied to the field. This type of problem can be solved in a variety of ways using, for example, the Arnoldi process in a software distribution called ARPACK. More details on Finite Difference Mode Solver can be found in [36, 37, 38, 39, 40].

A.2 FDTD Method

A complete description of the Finite Difference Time Domain (FDTD) method is presented in the book by A. Taflov *et al.* [41]. The essentials of the approach

APPENDIX A. SIMULATION TECHNIQUES

are reviewed here for completeness. The FDTD method represents a discretized implementation of Maxwell's equations. The fields are updated in time rather than space. The computational grid is arranged such that the divergence operations

$$\nabla \cdot \mathbf{D} = \rho \quad (\text{A.4})$$

and

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{A.5})$$

are naturally maintained. The electric and magnetic field components are offset from one another by 1/2 grid point so as to enable second order accuracy in the derivatives of Maxwell's equations.

The Finite Difference Time Domain method is based on a simple discretization of Maxwell's equations. We begin with

$$\nabla \times \mathbf{E} = -\mu \frac{\partial}{\partial t} \mathbf{H} \quad (\text{A.6})$$

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial}{\partial t} \mathbf{E} \quad (\text{A.7})$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \hat{x} \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) - \hat{y} \left(\frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z} \right) + \hat{z} \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \quad (\text{A.8})$$

APPENDIX A. SIMULATION TECHNIQUES

$$\frac{\partial}{\partial t} E_x = \frac{1}{\epsilon} \left(\frac{\partial}{\partial y} H_z - \frac{\partial}{\partial z} H_y \right) \quad (\text{A.9})$$

$$\frac{E_x|_{i+\frac{1}{2},j,k}^{n+1} - E_x|_{i+\frac{1}{2},j,k}^n}{\Delta t} = \frac{1}{\epsilon_{i,j,k}} \left(\frac{H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_z|_{i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_y|_{i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right) \quad (\text{A.10})$$

$$E_x|_{i+\frac{1}{2},j,k}^{n+1} = E_x|_{i+\frac{1}{2},j,k}^n + \frac{\Delta t}{\epsilon_{i,j,k}} \left(\frac{H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_z|_{i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_y|_{i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right) \quad (\text{A.11})$$

$$E_y|_{i,j+\frac{1}{2},k}^{n+1} = E_y|_{i,j+\frac{1}{2},k}^n + \frac{\Delta t}{\epsilon_{i,j,k}} \left(\frac{H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_z|_{i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_x|_{i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right) \quad (\text{A.12})$$

$$E_z|_{i,j,k+\frac{1}{2}}^{n+1} = E_z|_{i,j,k+\frac{1}{2}}^n + \frac{\Delta t}{\epsilon_{i,j,k}} \left(\frac{H_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_y|_{i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_x|_{i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right) \quad (\text{A.13})$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \hat{x} \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) - \hat{y} \left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \right) + \hat{z} \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \quad (\text{A.14})$$

$$\frac{\partial}{\partial t} H_x = -\frac{1}{\mu} \left(\frac{\partial}{\partial y} E_z - \frac{\partial}{\partial z} E_y \right) \quad (\text{A.15})$$

$$H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^{n-\frac{1}{2}} - \frac{\Delta t}{\mu_{i,j,k}} \left(\frac{E_z|_{i,j+1,k+\frac{1}{2}}^n - E_z|_{i,j,k+\frac{1}{2}}^n}{\Delta y} - \frac{E_y|_{i,j+\frac{1}{2},k+\frac{1}{2}}^n - E_y|_{i,j+\frac{1}{2},k-\frac{1}{2}}^n}{\Delta z} \right) \quad (\text{A.16})$$

$$H_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n-\frac{1}{2}} - \frac{\Delta t}{\mu_{i,j,k}} \left(\frac{E_z|_{i+1,j,k+\frac{1}{2}}^n - E_z|_{i,j,k+\frac{1}{2}}^n}{\Delta x} - \frac{E_x|_{i+\frac{1}{2},j,k+1}^n - E_x|_{i+\frac{1}{2},j,k}^n}{\Delta z} \right) \quad (\text{A.17})$$

$$H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n-\frac{1}{2}} - \frac{\Delta t}{\mu_{i,j,k}} \left(\frac{E_y|_{i+1,j+\frac{1}{2},k}^n - E_y|_{i,j+\frac{1}{2},k}^n}{\Delta x} - \frac{E_x|_{i+\frac{1}{2},j+1,k}^n - E_x|_{i+\frac{1}{2},j,k}^n}{\Delta y} \right) \quad (\text{A.18})$$

where i , j and k refer to the x , y and z grid points and n refers to the current time step. This grid is known as Yee grid [42] and is shown in Figure A.1.

As might be expected, as the discretizations in space and time approach zero, the algorithm becomes exact. However, clearly the use of finely spaced

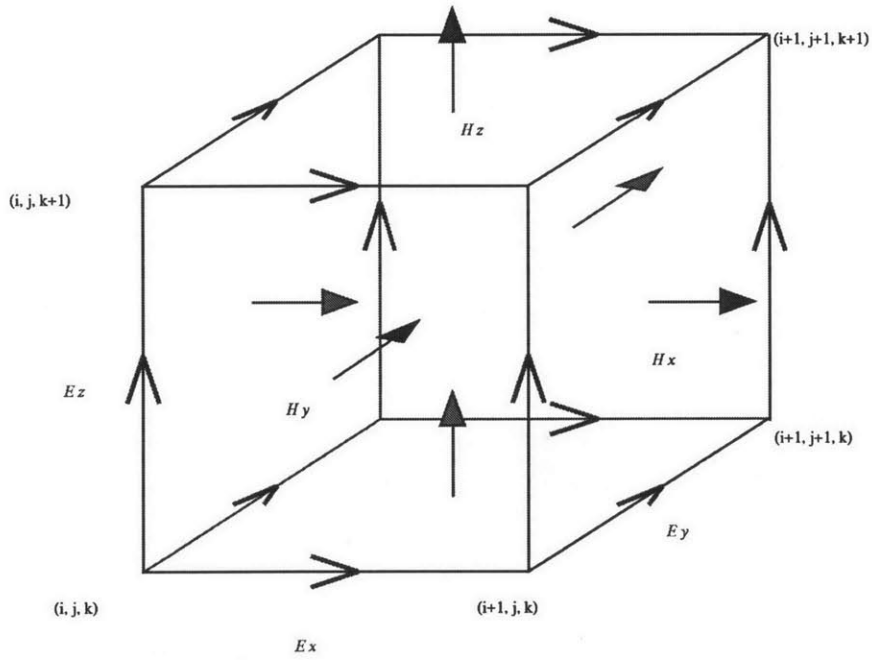


Figure A.1: Yee grid. Figure taken from http://ab-initio.mit.edu/wiki/index.php/Yee_grid

grid points increases the number of calculations required to update the field. Therefore, it is important to choose a fine, but not overly fine grid spacing. Taflove demonstrates that accurate results can be obtained with a spatial discretization of $\lambda/20$. The time step is then chosen to be close to $\Delta t = \Delta x / (c\sqrt{2})$. This value of Δt also represents the upper bound for stable operation of the algorithm. For time steps greater than Δt , the field grows without bound. And, for time steps less than Δt , numerical dispersion creeps into the propagating field. It is therefore desirable to choose a time step that is close to, but slightly less than Δt . A time step of $\Delta x / (2c)$ is generally sufficient. More details on Finite Difference Time Domain method can be found in [43, 42, 44, 45].

Appendix B

Source code

B.1 Modesolver code

This section contains MATLAB code that uses Mode Solver developed by Miloš Popović in Optics and Quantum Electronics research group at MIT. This code is using Mode Solver to solve for transverse eigenmodes and eigen wavevectors. Profiles of each mode are saved as .dat files and latter on used as input for the FDTD simulation.

```
% This script demonstrates how to use
%the mode solver from Milos Popovic.

close all; clear all;

addpath('C:\Users\Goran\ModeSolver');
```

APPENDIX B. SOURCE CODE

```
freq = 1.5:0.01:1.6;
```

```
n_Ge = 4.275;
```

```
n_Si = 3.47772;
```

```
n_SiO2 = 1.445;
```

```
n0 =1;
```

```
wavelength = 1.55; %um
```

```
k0 = 2*pi/wavelength;
```

```
n = [n0      n0      n0;  
      n0      n_Si    n0;  
      n_Si    n_Si    n_Si;  
      n_SiO2  n_SiO2  n_SiO2].';
```

```
% Ge Waveguide
```

```
Ge_W = 0.7; %um
```

```
Ge_H = 0.3; %um
```

```
%Si Waveguide
```

```
Rib_W = 1.0; %um
```

```
Rib_h = 0.2; %um
```

```
Si_H = 0.20; %um
```

APPENDIX B. SOURCE CODE

```
% Simulation Window
Window_W = 6.0; %um
Window_H = 3.5; %um

% SiO2 substrate
SiO_H = 1.0; %um

step_w = 0.025; %um
step_h = 0.025; %um

height = [(Window_W-Rib_W)/2 Rib_W (Window_W-Rib_W)/2];

width = [(Window_H-SiO_H-Si_H-Rib_h) Rib_h Si_H SiO_H];

% step size;
% steps that end on medium boudaries
% yield better precision

dxy = [step_w step_h];

% how many modes to calculate
options.NMODES_CALC = 1;
% left right bottom top;
options.PMLwidth = [0 0.0 0.0 0.0];
options.PMLsigma = [0.3 0.3];
options.mu_guess = k0*3.6; %guess for beta
```


APPENDIX B. SOURCE CODE

```
tic;
[N, F] = sisolver3d(n, width, height,...
                    dxy, k0, options);

toc; %for timing

F.beta

figure(1);
imagesc(F.Ey(:,:,3)); title('Ey');
imagesc(F.Ex(:,:,1)); title('Ex');

neff=F.beta/k0

figure(2);
imagesc(N.n); title('Refractive index profile',...
                  'FontSize',16);
set(gca,'YTick',[]);
set(gca,'XTick',[]);
colorbar;

for mode=1:options.NMODES_CALC
    figure;
    subplot(3,2,1); imagesc(abs(F.Ex(:,:,mode)));
    title('Ex');
    subplot(3,2,2); imagesc(abs(F.Ey(:,:,mode)));
    title('Ey');
    subplot(3,2,3); imagesc(abs(F.Ez(:,:,mode)));
```

```

    title('Ez');
    subplot(3,2,4); imagesc(abs(F.Hy(:, :, mode)));
    title('Hy');
    subplot(3,2,5); imagesc(abs(F.Hx(:, :, mode)));
    title('Hx');
    subplot(3,2,6); imagesc(abs(F.Hz(:, :, mode)));
    title('Hz');
end

SZ.F = F;
SZ.N = N;

fprintf('%s%d %d \n', 'size of F.Ex:', size(F.Ex));
fprintf('%s%d %d \n', 'size of F.Ey:', size(F.Ey));

save SZ
fixmode

figure(3);
imagesc(abs(F.Ey(:, :, 1)));
title('Mode profile', 'FontSize', 16);
set(gca, 'YTick', []);
set(gca, 'XTick', []);
colorbar;

```

B.2 3D FDTD code

This section contains MATLAB code that creates `input.in` file which is input for the 3D FDTD simulator. This simulator is developed and modified by members of Professor Michael Watts's research group at MIT.

```
% All units are in um

close all;  clear all;

Lx = 10;
Ly = 7;
Lz = 5;

Si_w = 5;
Si_h = 0.2;
Si_l = 8;

Rib_w = 1;
Rib_h = 0.2;

% File generation
fid = fopen('input.txt','wt');

% Space domain definition
dx = 0.05; dy = 0.05; dz = 0.05;
```

APPENDIX B. SOURCE CODE

```
cfl = 1.1;  %>1 for causation

% Grid size definition
% (dx, dy, dz, cfl, shift)
fprintf(fid, '%f %f %f %f %d\n', dx, dy, dz, cfl, 0);

% Simulation and PML range definition
% (max_x, max_y, max_z, PML_x, PML_y, PML_z)
PML_x = 16*dx; PML_y = 16*dy; PML_z = 16*dz;

%Lx = 20;
%Ly = 11.0;
%Lz = 2.2;

fprintf(fid, '%f %f %f %f %f %f\n', Lx, Ly, Lz, ...
        PML_x, PML_y, PML_z);

% Time domain definition
% (Tstart, Tend, Tstep, flag_snapshot)
Tstart = 0;  Tend = 20; %000;
Tstep = 10;  flag_snapshot = 0;

fprintf(fid, '%d %d %d %d\n', Tstart, Tend, ...
        Tstep, flag_snapshot);
```

APPENDIX B. SOURCE CODE

```
% Number of monitor slices
% Slice position: (0,0,z) or (0,y,0) or (x,0,0)
N_slices = 2;

fprintf(fid, '%d\n', N_slices);

fprintf(fid, '%f %f %f\n', 1.5, 0, 0);
fprintf(fid, '%f %f %f\n', 8.5, 0, 0);

%fprintf(fid, '%f %f %f\n', 0, 0, Lz/2);
%fprintf(fid, '%f %f %f\n', 0.5+dx, 0, 0);
%fprintf(fid, '%f %f %f\n', 1, 0, 0);
%fprintf(fid, '%f %f %f\n', 0, 0.5, 0);

% Dielectric objects
% (N_obj, background index)
N_obj = 2;
n0 = 1.445; %SiO2

fprintf(fid, '%d %f\n', N_obj, n0);

% Object 1
% (j_obj, number of j_obj)
% Rectangular waveguide
% (x_center, y_min, z_min, Lx, Ly, Lz, ...
%   side_slope, refractive index)
```

APPENDIX B. SOURCE CODE

```
n1 = 3.447; %Si
% waveguide position
%sw_pos = [Lx/2  2*dy+10  Lz/2];
x_off = 1;
y_off = 1;
oxide = 3;

fprintf(fid, '%d %d\n', 0, 1);
fprintf(fid, '%f %f %f %f %f %f %f %f\n', x_off+Si_l/2,...
      y_off,oxide, Si_l, Si_w, Si_h, 0, n1);

% (dir1(3), dir2(3), dir3(3))
% dir1 = [1 0 0];
% dir2 = [0 1 0];
% dir3 = [0 0 1];
fprintf(fid, '%f %f %f %f %f %f %f %f %f\n',...
      1, 0, 0, 0, 1, 0, 0, 0, 1);

% Object 2
% (j_obj, number of j_obj)
% Rectangular waveguide
% (x_center, y_min, z_min, Lx, Ly, Lz,...
%      side_slope, refractive index)
n_rib = 3.447; %Si
% waveguide position
%sw_pos = [Lx/2  2*dy+10  Lz/2];

fprintf(fid, '%d %d\n', 0, 1);
fprintf(fid, '%f %f %f %f %f %f %f %f\n',...
```

APPENDIX B. SOURCE CODE

```
x_off+Si_l/2,y_off+(Si_w-Rib_w)/2, oxide+Si_h,...
    Si_l, Rib_w, Rib_h, 0, n_rib);

% (dir1(3), dir2(3), dir3(3))
% dir1 = [1 0 0];
% dir2 = [0 1 0];
% dir3 = [0 0 1];
fprintf(fid, '%f %f %f %f %f %f %f %f %f\n',...
        1, 0, 0, 0, 1, 0, 0, 0, 1);

% Sources
% Number of sources
N_sources = 1;

fprintf(fid, '%d\n', N_sources);

% Source 1
% Source Time type: (1 for cw, 2 for pulse,...
%           lambda, pulse_width(fs), shift(pw))
fprintf(fid, '%d %f %f %f\n', 2, 1.55, 10, 3);

source_w = 101*dx;
source_h = 101*dz;

% Source Space type: (xmin, xmax, ymin, ymax, zmin, zmax)
```

APPENDIX B. SOURCE CODE

```
fprintf(fid, '%f %f %f %f %f %f\n', x_off, x_off, ...
        y_off+Si_w/2-source_w/2, y_off+Si_w/2+source_w/2, ...
        oxide+Si_h-source_h/2, oxide+Si_h+source_h/2);

% Source mode file: (neff, Ex, Hy, Ey, Hx)
neff = 3.1078; % from modesolver

fprintf(fid, '%f %s %s %s %s\n', neff, 'exr.dat', ...
        'hyr.dat', 'eyr.dat', 'hxr.dat');

% Source filed type: (jsource, pol, src)
% abs(jsource)=1, from mode solver: -1 backward, +1 forward
% pol: 0 horizontal, 1 vertical
% src: amplitude
fprintf(fid, '%d %d %f\n', 1, 0, 1);

% Output setup

% Frequency range
lambda_min = 1.50;
lambda_max = 1.60;
delta_lambda = 0.01;
Nfreq = (lambda_max-lambda_min)/delta_lambda;

fprintf(fid, '%f %f %f\n', lambda_min, ...
        lambda_max, delta_lambda);
```


APPENDIX B. SOURCE CODE

```
% Number of flux monitor slices
N_monitor = 2;

fprintf(fid, '%d\n', N_monitor);

% Flux monitor position: (x_min,x_max,y_min,y_max,z_min,z_max)

% Input
monly_min = y_off;
monly_max = y_off+Si_w;

monlz_min = oxide;
monlz_max = oxide+Si_h+Rib_h;

fprintf(fid, '%f %f %f %f %f %f\n', x_off+0.75,...
    x_off+0.75,monly_min, monly_max,...
    monlz_min, monlz_max);
fprintf(fid, '%d %d\n', Tstart, Tend);

% Thru
mon2y_min = y_off;
mon2y_max = y_off+Si_w;

mon2z_min = oxide;
mon2z_max = oxide+Si_h+Rib_h;

fprintf(fid, '%f %f %f %f %f %f\n', x_off+Si_l-1.25,...
    x_off+Si_l-1.25, mon2y_min, mon2y_max,...
```

APPENDIX B. SOURCE CODE

```
        mon2z_min, mon2z_max);
fprintf(fid, '%d %d\n', Tstart, Tend);

%fclose('input.in');

% Coarse calculation

dL = 1/sqrt(1/dx^2+1/dy^2+1/dz^2)/cfl;
ng = 4.3;

% Propagation distance is calculated by group velocity

fprintf(1, '%d %d %d\n', round(Lx/dx), ...
        round(Ly/dy), round(Lz/dz));
fprintf(1, 'Propagation length: x=%f um\n', Tend*dL/ng);
fprintf(1, 'Simulation window limit: x=%f\n', Lx);
fprintf(1, 'Critical x limit: x=%f\n', 6.5);
fprintf(1, 'May the force be with you!\n');

fclose(fid);
```

B.3 Meep code

This section contains code used by Meep. Meep is 3D FDTD simulator developed in Ab-Initio Physics Research Group at MIT to model electromagnetic systems. It is an alternative to simulator from previous section. Author's opinion is that Meep is more user friendly than 3D FDTD simulator from previous section. Also, it has many very useful built-in features for post processing. The (.ctl) file specifies the geometry we wish to study, the sources and the outputs we want to compute. This .ctl file is implemented as script in Scheme programming language. Here is included an example of control file (.ctl) that creates thick oxide, silicon waveguide on silica, absorbing germanium layer on top and computes fields in the structure. More details about Meep (and Scheme) can be found at Meep home page.

```
(define-param Lx 32)
(define-param Ly 16)
(define-param Ox_H 4)
(define-param Si_H 0.4)
(define-param Ge_W 8)
(define-param Ge_H 3)
(define-param PML 1)
(define-param x_off 2)

(define-param res 25)

(define-param n_Si 3.47772)
```

APPENDIX B. SOURCE CODE

```
(define-param n_Ge 4.275)
(define-param n_Ox 1.44402)

(set! geometry-lattice (make lattice (size Lx Ly no-size)))

(set! geometry (list
  (make block (center 0 0) (size (- Lx (* 2 PML)) Si_H infinity)
    (material (make dielectric (index n_Si))))
  (make block (center 0 (* -0.5 (+ Si_H Ox_H)))
    (size (- Lx (* 2 PML)) Ox_H infinity)
    (material (make dielectric (index n_Ox))))
  (make block (center (- (- (* 0.5 (- Lx Ge_W)) PML) x_off)
    (* 0.5 (+ Si_H Ge_H)))
    (size Ge_W Ge_H infinity)
    (material (make dielectric (index n_Ge))))))

(set! pml-layers (list (make pml (thickness PML))))
(set! resolution res)

; (set! sources (list
;   (make source
;     (src (make continuous-src
;       (wavelength (* 2 (sqrt 12))) (width 10)))
;     (component Ez)
;     (center -14.8 0) (size 0 0.1))))
```

APPENDIX B. SOURCE CODE

```
(define-param fcen 0.6452) ; pulse center frequency
(define-param df 0.3)      ; pulse width (in frequency)
(set! sources (list
  (make source
    (src (make gaussian-src (frequency fcen) (fwidth df)))
    component Ez)
    (center (+ 2 (* (- 0 Lx) 0.5)) 0)
    (size 0 Si.H))))

(define-param nfreq 100); number of freqs at which to compute flux

(define trans ; reflected flux
  (add-flux fcen df nfreq
    (make flux-region
      (center 12 2) (size 0 3))))

; (define trans ; transmitted flux
;   (add-flux fcen df nfreq
```

APPENDIX B. SOURCE CODE

```
;          (if no-bend?
;          (make flux-region
;          (center (- (/ sx 2) 1.5) wvg-ycen) (size 0 (* w 2)))
;          (make flux-region
;          (center wvg-xcen (- (/ sy 2) 1.5)) (size (* w 2) 0))))

(define refl ; reflected flux
  (add-flux fcen df nfreq
    (make flux-region
      (center 13 1.5) (size 0 4))))

(run-until 200
  (at-beginning output-epsilon)
  (to-appended "ez" (at-every 0.6 output-efield-z)))

(display-fluxes trans refl)
```


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